Tensor Calculus

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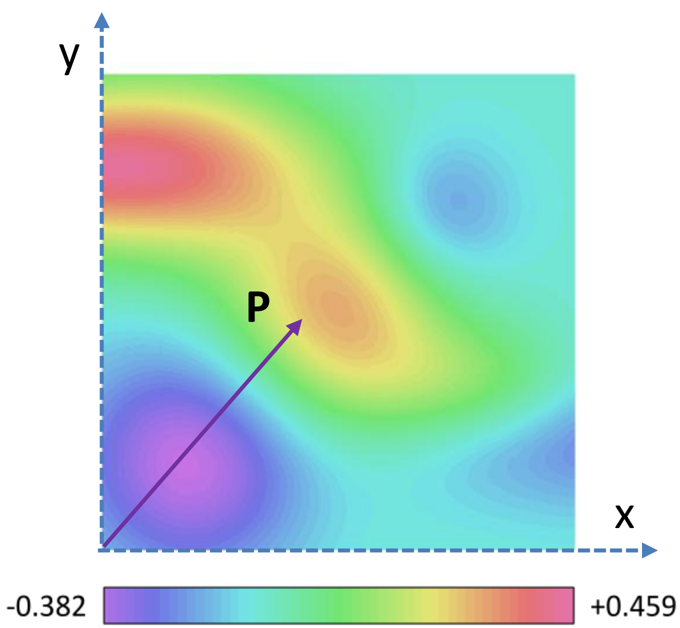
Scalar Function

Let **P** be a position vector in a Vector Space **V** with the basis of {$\mathbf{x}_1$, $\mathbf{x}_2$, … , $\mathbf{x}_n$ }, and *f*(**P**) is a function *f*: **V** ⟶ R. Then *f*(**P**) is Scalar Function, which is commonly known as **Scalar Field**.

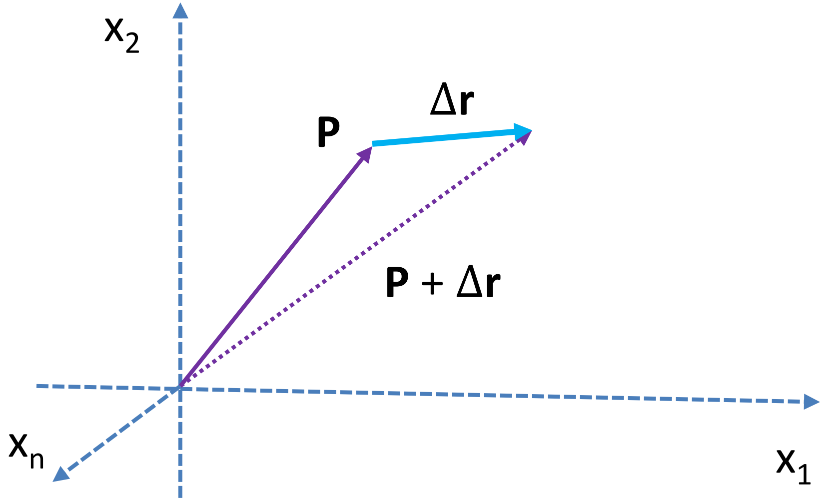
For example, we can visualize it as **Temperature** in a Region, *f*(**x**, **y**).



f(**x**, **y**)

Directional Derivative

Now, let **P** be a position vector in a Vector Space **V** with the basis of {$\mathbf{x}_1$, $\mathbf{x}_2$, … , $\mathbf{x}_n$ }, and *f*(**P**) is a function *f*: **V** ⟶ R. Then *f*(**P** + Δ**r**) is also defined around the neighborhood of **P** in **r** direction.

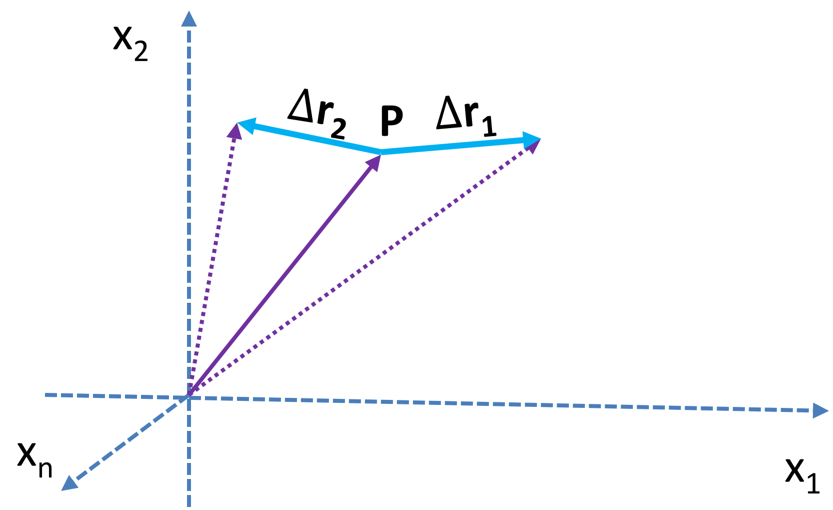


Now if we take the limit of Δ**r** to zero, then the $\frac{df}{d\boldsymbol{r}}$ is the **directional derivative** at **P** in the direction of **r**:

$$\lim_{\Delta \boldsymbol{r} \to 0} \frac{f(\boldsymbol{P} + \Delta \boldsymbol{r}) - f(\boldsymbol{P})}{\Delta \boldsymbol{r}} = \frac{df}{d\boldsymbol{r}}(\boldsymbol{P})$$

Tangents

However, as we can see, we can have different Δ**r** in different directions, and thus, we can have different directional derivatives of f(**P**) around the neighborhood of **P**.



So, collectively, all the different directional derivatives of f(**P**) creates a collection of **Tangents** around **P**. And f(**P**) is defined as **C^1 Smooth** around its neighborhood, as its **first derivative** is defined in any direction around the Neighborhood of **P**. If all of its derivatives are defined around the Neighborhood of **P**, then f(**P**) is known as **C^∞ Smooth**.

Partial Derivatives

So, as we can see, $\frac{df}{d\boldsymbol{r}}$ is the directional derivative of *f*(**P**) in the direction of **r**.

However, we can define the differential of *f*(**P**), without any reference to direction of **r**. In order to do so, we have to define the **Vector Space of Tangents** around the neighborhood of **P**. But what is **the basis** of such **Tangent Space** at P?

Since, Δ**r** is a vector around the neighborhood of **P**, we can define it as a Linear Combination of Δ of **Unit Coordinates**, *parallel transported* from the **Origin** to **P**, such that:

$$\Delta\boldsymbol{r} = \Delta\boldsymbol{x_1} + \cdots + \Delta\boldsymbol{x_n} = \sum_{i=1}^{n} \Delta\boldsymbol{x_i}$$

However, if we parameterize **r** with **r**(t), then:

$$\Delta\boldsymbol{r} = \boldsymbol{r}(t+h) - \boldsymbol{r}(t)$$

Therefore,

$$\Delta\boldsymbol{x_i} = \boldsymbol{x_i}(t+h) - \boldsymbol{x_i}(t)$$

Partial Derivatives

As we can see,

$$\frac{df}{d\boldsymbol{r}(t)} = \frac{\partial f}{\partial \boldsymbol{x_1}}\frac{d\boldsymbol{x_1}}{dt} + \cdots + \frac{\partial f}{\partial \boldsymbol{x_n}}\frac{d\boldsymbol{x_n}}{dt}$$

, where each component is defined as:

$$\lim_{\boldsymbol{h}\to 0}\frac{f\big(\boldsymbol{P} + \boldsymbol{x_i}(t+h)\big) - f(\boldsymbol{P} + \boldsymbol{x_i}(t))}{\boldsymbol{x_i}(t+h) - \boldsymbol{x_i}(t)} \cdot \lim_{h\to 0}\frac{\boldsymbol{x_i}(t+h) - \boldsymbol{x_i}(t)}{h}$$

Then, the **directional derivative** at **P** can be rewritten as:

$$\frac{df}{d\boldsymbol{r}(t)} = \frac{df}{d\boldsymbol{r}}\frac{d\boldsymbol{r}}{dt} = \sum_{i=1}^{n}\frac{\partial}{\partial \boldsymbol{x_i}}\frac{d\boldsymbol{x_i}}{dt} f(\boldsymbol{P})$$

Tangent Space

As we can see, any directional derivative can be written as a **Linear Combination of Partial Derivatives**, without any reference to **r**. More intuitively, with the abuse of notation, it looks like a **Inner Product** of the **Partial Derivatives** and **Directional Vector**.

$$\frac{\partial f}{\partial \boldsymbol{r}(t)} = \frac{\partial f}{\partial \boldsymbol{r}} \cdot \frac{d\boldsymbol{r}}{dt} = \sum_{i=1}^{n} \frac{\partial}{\partial \boldsymbol{x_i}} f(\boldsymbol{P}) \cdot \frac{d\boldsymbol{x_i}}{dt}$$

At this point, we can see that any vector **u** in **Tangent Space U** at **P** could possibly be the **Linear Span of Partial Derivatives** as its basis such that:

$$\boldsymbol{u}(\boldsymbol{P}) = \sum_{i=1}^{n} g_i(\boldsymbol{P}) \frac{\partial}{\partial \boldsymbol{x_i}}$$

As we can see, although **Tangent Space U** has the same dimension as **Vector Space V**, their bases are different.

Tangent Space

Formally, we can define **Tangent Vector v** $\in \mathbf{T}_p$ at **p** $\in \mathbf{V}$ with the basis of $\{\mathbf{x}_1, \dots, \mathbf{x}_n\}$ as a set of pairs:

$$T_p\boldsymbol{R}^n \to \{(\boldsymbol{p}, \boldsymbol{v});\ \boldsymbol{v} \in \boldsymbol{U}, \boldsymbol{p} \in \boldsymbol{V}\}$$

or

$$(\boldsymbol{p}, \boldsymbol{v}) \to \boldsymbol{v}: \boldsymbol{v}(\boldsymbol{p})$$

Tangent Space $T_p\boldsymbol{R}^n$ is a Vector Space, with the basis $\left\{\frac{\partial}{\partial \boldsymbol{x}_1}, \dots, \frac{\partial}{\partial \boldsymbol{x}_n}\right\}$, such that:

$$\boldsymbol{v}(\boldsymbol{x}_i) = g_i(\boldsymbol{x}_i)\frac{\partial}{\partial \boldsymbol{x}_i}$$

, where $\mathbf{x}_i$ is the basis of **V** and $\partial/\partial\mathbf{x}_i$ is the basis of $\mathbf{T}_p$.

As we can see, **g**($\mathbf{x}_i$) is the Metric Tensor, which defines the map between the two corresponding bases of **V** and $\mathbf{T}_p$, but in **Rectilinear** Coordinates, such Metric Tensor is **Krönecker Delta**.

Tangent Bundle

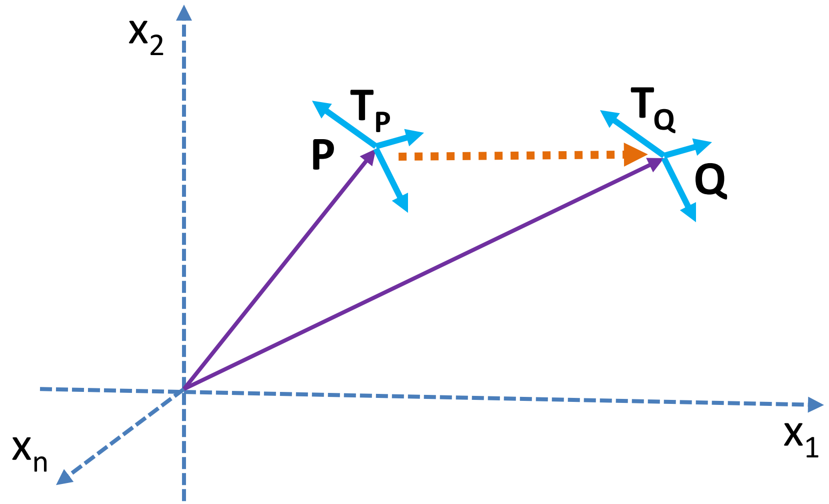
However, as we can see, the Tangent Space $\mathbf{T_p}$ is still at point **P** in the vector space **V**.

As we know, there are many points $\{\mathbf{P}_1, \mathbf{P}_2, \ldots\}$ in the vector space **V**, and each point $\mathbf{P}_i$ has its own corresponding **Tangent Space** $\mathbf{T_{pi}}$.

The collection of all the **Tangent Space** of every point in vector space **V** is known as **Tangent Bundle TP**.

Parallel Transport

As we can see, each different point $\mathbf{P_i}$ in vector space **V** has different Tangent Space $\mathbf{T_{pi}}$.



At this point, it should naturally occur to us as to how each **Tangent Space** $\mathbf{T_p}$ connects to each other. For now, we will just assume that the whole **Tangent Bundle** is $\mathbf{C^\infty}$ **Smooth**, and each **Tangent Space** $\mathbf{T_p}$ is **continuously connected** to each other.

But remember, it is not always the case, and for such cases, we need to define the **Parallel Transport** for those **Connections**. One of the famous connections is **Levi-Civita Connection**.

Vector Fields

Previously, we are aware that **Tangent Space T_p** at **P** is defined as:

$$\boldsymbol{v}(\boldsymbol{P}) = \sum_{i=1}^{n} g_i(\boldsymbol{P}) \frac{\partial}{\partial \boldsymbol{x_i}}$$

So, if we have every **Tangent Space T_p** in anywhere **x** in **Tangent Bundle**, then **Vector Field** can be defined as:

$$\boldsymbol{v_p}(x) = \sum_{i=1}^{n} g_i(x) \frac{\partial}{\partial \boldsymbol{x_i}}$$

Vector Fields

When it comes to Vector Fields, we have to be aware of 2 important aspects.

First, **Tangent Space T_p** exists differently from the Vector Space **V**, with different bases. Therefore, we always have to define **Coordinate Transformation**. Fortunately, in Rectilinear Coordinate Systems, such as Euclidean Coordinates, such Transformation is **Krönecker Delta**.

$$g_t : R^n \to T_p R^n$$

Second, each **Tangent Space T_p** exists differently from other **Tangent Space T_p** in the **Tangent Bundle**. Therefore, we also need to define **Connections** between **Tangent Spaces,** known as **Parallel Transport**, which is defined by **Inner Product Space**. Again, in Rectilinear Coordinate Systems, we can have Smooth Connections, without any changes.

$$g_c : T_p R^n \times T_p R^n \to R$$

Metric Tensor

As we know, **Metric Tensor** is defined by the **Inner Product** between two **bases of vectors**, such that:

$$g_{ij} = g_i \cdot g_j$$

Likewise, we can define Metric Tensor between the bases of Vector Fields, such that:

$$g\colon T_p R^n \times T_q R^m \to R$$

$$g_{ij} = \langle \frac{\partial}{\partial x_i}, \frac{\partial}{\partial x_j} \rangle = \frac{\partial}{\partial x_i} \cdot \frac{\partial}{\partial x_j}$$

Metric Tensor

Since, the inner product of the two bases is:

$$g_{ij} = \langle \frac{\partial}{\partial x_i}, \frac{\partial}{\partial x_j} \rangle = \frac{\partial}{\partial x_i} \cdot \frac{\partial}{\partial x_j}$$

In Matrix Form, we will have:

$$g_{ij} = \begin{pmatrix} \frac{\partial}{\partial \boldsymbol{x_1}} \frac{\partial}{\partial \boldsymbol{x_1}} & \cdots & \frac{\partial}{\partial \boldsymbol{x_1}} \frac{\partial}{\partial \boldsymbol{x_m}} \\ \vdots & \ddots & \vdots \\ \frac{\partial}{\partial \boldsymbol{x_n}} \frac{\partial}{\partial \boldsymbol{x_1}} & \cdots & \frac{\partial}{\partial \boldsymbol{x_n}} \frac{\partial}{\partial \boldsymbol{x_m}} \end{pmatrix}$$

Coordinate Transformation

Now, we will define the Coordinate Transform between the Vector Space **V** , with the basis { $\mathbf{x_1}$, …, $\mathbf{x_n}$ } and the Tangent Space $\mathbf{T_p}$, with the basis {$\partial/\partial\mathbf{x}_1$, …, $\partial/\partial\mathbf{x}_n$ }.

Thus,

$$g_{ij} = \langle\frac{\partial}{\partial \boldsymbol{x}_i}, \boldsymbol{x}_j\rangle = \frac{\partial}{\partial \boldsymbol{x}_i}\boldsymbol{x}_j = \delta_{ij}$$

Now, if we try to transform the scalar function *f*(**x**) from the Vector Space **V** to the Tangent Space $\mathbf{T_p}$, then:

$$\langle\frac{\partial}{\partial \boldsymbol{x}_i}, f(\boldsymbol{x})\boldsymbol{x}_j\rangle = \frac{\partial}{\partial \boldsymbol{x}_i} f(\boldsymbol{x})\frac{\partial}{\partial \boldsymbol{x}_i}\boldsymbol{x}_j = \frac{\partial f}{\partial \boldsymbol{x}_i}\delta_{ij}$$

Directional Derivative

With the definition of Vector Field, we will redefine the **Directional Derivative** more generally.

Let $f(\mathbf{x})$ be a differentiable Scalar Function on **V** at $\mathbf{p} \in \mathbf{V}$, and if **v** be a Tangent Vector in Tangent Space $\mathbf{T_p}$, then **Directional Derivative** $D_v(f(x))$ can be defined as an **Inner Product**:

$$D_{\boldsymbol{v}}\big(f(x)\big) = \boldsymbol{v_p}\big(f(x)\big) \cdot \boldsymbol{v_p}(\boldsymbol{v})$$

However,

$$\boldsymbol{v_p}(x) = \sum_{i=1}^{n} g_i(x) \frac{\partial}{\partial \boldsymbol{x_i}}$$

Thus,

$$\boldsymbol{v_p}(f(\boldsymbol{x})) = \sum_{i=1}^{n} g_i(f(x)) \frac{\partial}{\partial \boldsymbol{x_i}} \quad \boldsymbol{v_p}(\boldsymbol{v}) = \sum_{i=1}^{n} g_i(\boldsymbol{v}) \frac{\partial}{\partial \boldsymbol{x_i}} = \sum_{i=1}^{n} v_i \frac{\partial}{\partial \boldsymbol{x_i}}$$

Directional Derivative

But g_i is the **Coordinate Transformation**, which is, for now, a **Gram Matrix**, and $f(\mathbf{x})$ is a scalar, and so:

$$g_i(f(x)) = T^{ij}\frac{\partial f}{\partial \boldsymbol{x_i}} \qquad T^{ij} = \delta^{ij}, for\ R^n$$

Therefore,

$$\boldsymbol{v_p}(f(\boldsymbol{x})) = \sum_{i=1}^{n} T^{ij}\frac{\partial f}{\partial \boldsymbol{x_i}}\frac{\partial}{\partial \boldsymbol{x_i}}$$

Thus,

$$D_{\boldsymbol{v}}\big(f(x)\big) = \sum_{j=1}^{n} T^{ij}\frac{\partial f}{\partial \boldsymbol{x_i}} v_i$$

Gradient

Now, we can define Gradient of Scalar Function $f(\mathbf{x})$ as $\boldsymbol{\nabla} f(\mathbf{x})$ such that:

$$\nabla f(x) = \sum_{i=1}^{n} T^{ij} \frac{\partial f}{\partial \boldsymbol{x_i}} \frac{\partial}{\partial \boldsymbol{x_i}}$$

With Gradient Field, we can also denote Directional Derivative as:

$$D_{\boldsymbol{v}}\big(f(x)\big) = \nabla_{\boldsymbol{v}} f(\boldsymbol{x}) = \nabla f(\boldsymbol{x}).\boldsymbol{v} = \sum_{i=1}^{n} T^{ij} \frac{\partial f}{\partial \boldsymbol{x_i}} v_i$$

Derivative Operations

Now, Directional Derivative, $\nabla_v f(\mathbf{x})$ can be defined as Derivative Operations, which have the following properties. We will use this notation ∇_v for later definitions, instead of D_v.

$$D_{\boldsymbol{v}}\big(f(x)\big) = \nabla_{\boldsymbol{v}} f(\boldsymbol{x}) = \nabla f(\boldsymbol{x}).\boldsymbol{v}$$

$$\nabla_{\mathrm{v}}(f+\mathrm{g}) = \nabla_{\mathrm{v}}(f) + \nabla_{\mathrm{v}}(\mathrm{g})$$

$$\nabla_{\mathrm{v}}(cf) = \mathrm{c}\,\nabla_{\mathrm{v}}(f)$$

$$\nabla_{\mathrm{v}}(f\mathrm{g}) = \mathrm{g}\,\nabla_{\mathrm{v}}(f) + f\,\nabla_{\mathrm{v}}(\mathrm{g})$$

$$\nabla_{\mathrm{v}}(f \circ \mathrm{g}) = \partial(f(\mathrm{g}))\nabla_{\mathrm{v}}(\mathrm{g})$$

Basis of Vector Fields

As we know, **Tangent Space** $T_p \boldsymbol{R}^n$ is a Vector Space, with the **basis** $\left\{ \frac{\partial}{\partial \boldsymbol{x_1}}, \cdots, \frac{\partial}{\partial \boldsymbol{x_n}} \right\}$.

However, such notation is a little bit overwhelming, and therefore, we can simply denote the **basis** of Vector Field, or Tangent Space, as { $\boldsymbol{\partial}_1$, ..., $\boldsymbol{\partial}_{\mathrm{n}}$ }.

With this new notation, it is a lot simpler to represent the Vector Field more intuitively.

Covariant Derivative

Now, at this point, it should naturally occur to us as to what if *f*(**x**) is a Vector Function, instead of a Scalar Function, such as: $f(\boldsymbol{x}): R^n \to R$

In fact, if *f*(**x**) is a Vector Function : $f(\boldsymbol{x}): R^n \to R^m$, we can actually assume it as another **Vector Field W**.

With this in mind, we can now define **Covariant Derivative**, which is a **Extension** of Directional Derivative for any *f*(**x**):

$$f(\boldsymbol{x}): R^n \to R \quad \text{or} \quad f(\boldsymbol{x}): R^n \to R^m$$

Covariant Derivative can be define as the derivative of the **Connection** between two Vector Fields, namely Vector Function **W**: *f*(**x**) and Tangent Space **V**: $\mathbf{T_p}$ such that:

$$\nabla_v W = \nabla_{v^i \boldsymbol{\partial}_i}(w^j \boldsymbol{\partial}_j)$$

Covariant Derivative

More formally, Covariant Derivative can be defined as **Connection** on the **Tangent Bundle** to other **Tangent Bundles**.

Suppose a Vector Field **W** is a Vector Function with the basis { $\boldsymbol{\partial}_1$, …, $\boldsymbol{\partial}_k$ } and Tangent Space **V**: $\mathbf{T_p}$ with the basis { $\boldsymbol{\partial}_1$, …, $\boldsymbol{\partial}_n$ }.

Then, we will define the **Metric** (**Coordinate**) **Connection** between the **Tangent Bundles** such that:

$$\nabla_{\boldsymbol{\partial}_i}\boldsymbol{\partial}_j = \Gamma_{ij}^{k}\boldsymbol{\partial}_k \text{ , where } \Gamma_{ij}^{k} = \text{Christoffel Symbols}$$

We will look into Christoffel Symbols later, for now, let us continue to define Covariant Derivatives as **Interior Product** of **W** with **V**.

$$\nabla_v W = \nabla_{v^i \boldsymbol{\partial}_i}(w^j \boldsymbol{\partial}_j)$$

Covariant Derivative

As we can see,

$$\nabla_v W = \nabla_{v^i \boldsymbol{\partial}_i}(w^j \boldsymbol{\partial}_j)$$

$$= v^i \nabla_{\boldsymbol{\partial}_i}(w^j \boldsymbol{\partial}_j)$$

By Leibniz Rule

$$= v^i w^j \nabla_{\boldsymbol{\partial}_i} \boldsymbol{\partial}_j + v^i \boldsymbol{\partial}_j \nabla_{\boldsymbol{\partial}_i}(w^j)$$

But $\nabla_{\boldsymbol{\partial}_i}(w^j) = \dfrac{\partial w^j}{\partial \boldsymbol{x}_i}$

$$= v^i w^j \Gamma_{ij}^k \boldsymbol{\partial}_k + v^i \frac{\partial w^j}{\partial \boldsymbol{x}_i} \boldsymbol{\partial}_j$$

Therefore,

$$\nabla_v W = \left(v^i w^j \Gamma_{ij}^k + v^i \frac{\partial w^j}{\partial \boldsymbol{x}_i} \right) \boldsymbol{\partial}_k$$

Christoffel Symbols

As we already know, the Metric Tensor between two bases of Vector Fields is defined by:

$$g_{ij} = \langle \frac{\partial}{\partial x_i}, \frac{\partial}{\partial x_j} \rangle = \frac{\partial}{\partial x_i} \cdot \frac{\partial}{\partial x_j}$$

With this in mind, we will try to define Christoffel Symbols.

Suppose a Vector Field **W** is a Vector Function with the basis { $\boldsymbol{\partial}_1$, ..., $\boldsymbol{\partial}_k$ } and Tangent Space **V**: $\mathbf{T_p}$ with the basis { $\boldsymbol{\partial}_1$, ..., $\boldsymbol{\partial}_n$ }. If we try to transform a vector **w** from Vector Field **ψ** to Tangent Space, then:

$$\langle \frac{\partial}{\partial x_i}, w^j \frac{\partial \psi}{\partial x_j} \rangle = \frac{\partial}{\partial x_i} \left(w^j \frac{\partial \psi}{\partial x_j} \right)$$

Christoffel Symbols

Since

$$\langle \frac{\partial}{\partial x_i}, w^j \frac{\partial \psi}{\partial x_j} \rangle = \frac{\partial}{\partial x_i}\left(w^j \frac{\partial \psi}{\partial x_j} \right)$$

$$= \frac{\partial w^j}{\partial x_i}\frac{\partial \psi}{\partial x_j} + w^j \frac{\partial^2 \psi}{\partial x_i \partial x_j}$$

But $w^j \dfrac{\partial^2 \psi}{\partial x_i \partial x_j} = w^j \Gamma_{ij}^k \dfrac{\partial \psi}{\partial x_k} + L_{ij}\vec{n}$

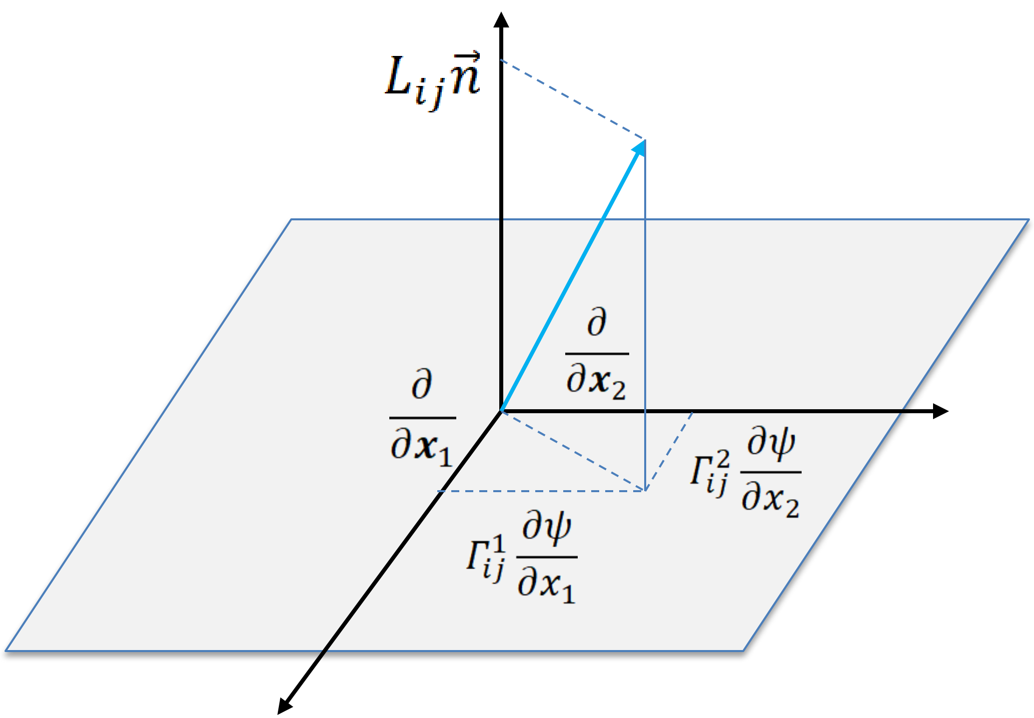
$$= \frac{\partial w^j}{\partial x_i}\frac{\partial \psi}{\partial x_j} + w^j \Gamma_{ij}^k \frac{\partial \psi}{\partial x_k} + L_{ij}\vec{n}$$

$$= \left(\frac{\partial w^k}{\partial x_i} + w^j \Gamma_{ij}^k \right) \frac{\partial \psi}{\partial x_k} + L_{ij}\vec{n}$$

Christoffel Symbols

We can visualize

$$w^j \frac{\partial^2 \psi}{\partial x_i \partial x_j} = w^j \Gamma_{ij}^k \frac{\partial \psi}{\partial x_k} + L_{ij}\vec{n}$$



Christoffel Symbols

As we can see,

$$\frac{\partial}{\partial x_i}\left(w^j \frac{\partial \psi}{\partial x_j}\right) - L_{ij}\vec{n} = \left(\frac{\partial w^k}{\partial x_i} + w^j \Gamma_{ij}^k\right)\frac{\partial \psi}{\partial x_k}$$

Again,

$$g_{ij} = \langle \frac{\partial \psi}{\partial x_i}, \frac{\partial \psi}{\partial x_j} \rangle = \frac{\partial^2 \psi}{\partial x_i \partial x_j}$$

$$\langle g_{ij}, g_l \rangle = \langle \frac{\partial^2 \psi}{\partial x_i \partial x_j}, \frac{\partial \psi}{\partial x_l} \rangle = \langle \Gamma_{ij}^k \frac{\partial \psi}{\partial x_k} + L_{ij}\vec{n}, \frac{\partial \psi}{\partial x_l} \rangle$$

$$= \langle \frac{\partial \psi}{\partial x_k}, \frac{\partial \psi}{\partial x_l} \rangle \Gamma_{ij}^k$$

$$= g_{kl}\Gamma_{ij}^k$$

Christoffel Symbols

Later, we will look into more as to how Christoffel Symbols is related to **Connections**, such as **Levi-Civita Connections**.

For now, we can assume that Christoffel Symbols are **Tangential Components** of **Second Order Derivatives** of **Vector Fields**.

As we know, **Inner Product** defines the Projection of one vector to another, and therefore, **Interior Product** of Vector Fields represents the similar Projection, and Christoffel Symbols represents **Tangential Components** of that Projection onto another Vector Field.

We can understand why **Tangential Components** would be particularly important to connect different Vector Bundles, and we can think of them as **Correction Terms** for Connections.

Differential Map

At this point, it should naturally occur to us as to what really is **Tensor Field**. Towards this end, let us start with the Differential Map.

As we know already, we can define a Linear Map **A**: **V** $\longrightarrow$ **U**, between two Vector Spaces as Vector Space **Homomorphism**.

Now, we will define a map between two Tangent Spaces such that **J**: $\mathbf{T_x R^n} \longrightarrow \mathbf{T_{f(x)} R^m}$, which is a map between two Vector Fields, known as **Differential Map** or **Diffeomorphism**, and it is represented by **Jacobian Matrix**.

The Differential Map, also known as **Pushforward**, is defined by *Linear Approximation* of Non-Linear function *f*(**x**) around the region of point **x** and its image *f*(**x**), such that:

$$L_{x_0}(x) = f(x_0) + f'(x - x_0)$$

Differential Map

Let F:***V*** ⟶ ***W*** be a nonlinear map between Vector Space ***V*** and ***W*** of dimension ***n*** and ***m***, respectively. Let { $\mathbf{x}_i$ } and { $\mathbf{y}_j$ } be bases of ***V*** and ***W***, respectively.

If we can define **basis maps** ϕ_V:***V*** ⟶ R^n and ϕ_W:***W*** ⟶ R^m for any **v** ∈ ***V*** and **w** ∈ ***W***, such that:

$$\phi_v(\boldsymbol{v}) = \phi_v\left(\sum_{i=1}^{n} v_i \boldsymbol{x}_i\right) \qquad \phi_w(\mathbf{w}) = \phi_w\left(\sum_{j=1}^{m} w_j \boldsymbol{y}_j\right)$$

Then, the function F:***V*** ⟶ ***W*** can be defined as the map $\phi_W \circ F \circ \phi^{-1}_V : R^n \longrightarrow R^m$.

$$\phi_w(w) = F\big(\phi_v(v)\big)$$

As we can see,

$$\phi'_w(w) = DF\big(\phi_v(v)\big)\phi_v{}'(v)$$

Differential Map

The Fréchet derivative *D*F of the nonlinear map F provides a *locally linear* approximation of F. The **linearization** of F at $x_0 \in U \subseteq R^n$ is defined as the map $L_{x0}: R^n \longrightarrow R^m$ defined as:

$$L_{x_0}(x) = F(x_0) + D_{x_0}F(x - x_0)$$

As we can see, around the region of $x_0 \in U \subseteq R^n$, $F_{x0}(x)$ can be approximated by **Differential Map** $D_{x0}F$. Thus, $D_{x0}F \in \mathrm{L}(R^n, R^m)$ is approximately a **Linear Map** near the region of $x_0 \in U \subseteq R^n$.

Since $\mathbf{F} = (F_1, \ldots, F_m)$ where each component of $F_i: U \longrightarrow R$, then $D_{x0}F$ can be represented as:

$$D_{x_0}F = \partial_j F_i(x_0)$$

Basis of Differential Map

As we can see, the **Differential Map**, or more appropriately **Jacobian Matrix**, can be defined as:

$$D_{x_0}F = \partial_j F_i(x_0)$$

Now, we will look into the basis of Jacobian Matrix. With the basis of $\mathbf{x_i}$ and $\mathbf{y_j}$:

$$(y_j \otimes x_i) D_{x_0}F = y_j \cdot D_{x_0}F(x_i) = \partial_j F_i(x_0)$$

Therefore,

$$D_{x_0}F = \partial_j F_i(x_0)\, y_j \otimes x_i$$

Jacobian Matrix

We can denote Jacobian Matrix as:

$$D_{x_0}F = \partial_j F_i(x_0)\, y_j \otimes x_i$$

Or more simply as:

$$J = \partial_j F_i = \frac{\partial F_j}{\partial x_i}$$

In Matrix Form,

$$J = \begin{pmatrix} \dfrac{\partial F_1}{\partial \boldsymbol{x_1}} & \cdots & \dfrac{\partial F_m}{\partial \boldsymbol{x_1}} \\ \vdots & \ddots & \vdots \\ \dfrac{\partial F_1}{\partial \boldsymbol{x_n}} & \cdots & \dfrac{\partial F_m}{\partial \boldsymbol{x_n}} \end{pmatrix}$$

Basis of Tangent Space

As we know already, if **v** be a Tangent Vector in Tangent Space $\mathbf{T_p}$, then we can define:

$$\boldsymbol{v_p}(x) = \sum_{i=1}^{n} g_i(x) \frac{\partial}{\partial \boldsymbol{x_i}}$$

The basis of Tangent Space $\mathbf{T_p}$ is usually represented by differentials:

$$\{\partial_1, \cdots, \partial_n\} = \left\{\frac{\partial}{\partial \boldsymbol{x_1}}, \cdots, \frac{\partial}{\partial \boldsymbol{x_n}}\right\}$$

However, as we know, the Coordinates of Tangent Space $\mathbf{T_p}$ can have different bases, such as **Rectilinear Coordinates** or **Curvilinear Coordinates**, depending on the choice of basis.

Reciprocal Basis of Tangent Space

We will also define the reciprocal basis of Tangent Space. Suppose, the basis of Tangent Space $\mathbf{T_p}$ is defined by:

$$\{\partial_1, \cdots, \partial_n\} = \left\{\frac{\partial \psi}{\partial \boldsymbol{x_1}}, \cdots, \frac{\partial \psi}{\partial \boldsymbol{x_n}}\right\}$$

Then, the reciprocal basis of Tangent Space can be defined by:

$$\delta_j^i = \sum_{i=1}^{n} \frac{\partial \boldsymbol{x}_i}{\partial \psi} \frac{\partial \psi}{\partial \boldsymbol{x}_j}$$

Inner Product of Tangent Space

Now we will define the Inner Product Space of Tangent Space $\mathbf{T_p}$ can be defined by Metric Tensor:

$$g_{ij} = g_i \cdot g_j$$

Since the basis of Tangent Space $\mathbf{T_p}$ is { $\partial/\partial\mathbf{x}_1$, ..., $\partial/\partial\mathbf{x}_\mathbf{n}$ }, and thus

$$g_{ij} = \langle \frac{\partial}{\partial x_i}, \frac{\partial}{\partial x_j} \rangle = \frac{\partial}{\partial x_i} \cdot \frac{\partial}{\partial x_j}$$

However, as we know, Inner Product Space is not always orthogonal, and therefore, we have to explicitly define **Orthogonality** of Tangent Spaces.

Inner Product of Tangent Space

As we can see, since the Inner Product Space defined by **Metric Tensor** can be represented as

$$g_{ij} = \langle \frac{\partial}{\partial x_i}, \frac{\partial}{\partial x_j} \rangle = \frac{\partial}{\partial x_i} \cdot \frac{\partial}{\partial x_j}$$

Then, let **ψ** be a Vector Field with the basis { $\boldsymbol{\partial}_1$, ..., $\boldsymbol{\partial}_n$ }. Then, the following can be defined as:

$$\langle \frac{\partial \psi}{\partial x_i}, \frac{\partial \psi}{\partial x_j} \rangle = \begin{pmatrix} \frac{\partial^2 \psi_1}{\partial \boldsymbol{x}_1^2} & \cdots & \frac{\partial^2 \psi_n}{\partial \boldsymbol{x}_1 \partial \boldsymbol{x}_n} \\ \vdots & \ddots & \vdots \\ \frac{\partial^2 \psi_1}{\partial \boldsymbol{x}_n \partial \boldsymbol{x}_1} & \cdots & \frac{\partial^2 \psi_n}{\partial \boldsymbol{x}_n^2} \end{pmatrix}$$

Orthongonality of Tangent Space

In order to define **Orthogonality** of Tangent Spaces, we have to define the following condition, without which **Orthogonality** cannot exist.

$$g_{ij} = \langle \frac{\partial}{\partial x_i}, \frac{\partial}{\partial x_j} \rangle = g_{ji} = \langle \frac{\partial}{\partial x_j}, \frac{\partial}{\partial x_i} \rangle = \delta_{ji}$$

Let $\boldsymbol{\psi}$ be a Vector Field with the basis { $\boldsymbol{\partial}_1$, ..., $\boldsymbol{\partial}_\mathrm{n}$ }. Then, the following can be defined as:

$$\langle \frac{\partial \psi}{\partial x_i}, \frac{\partial \psi}{\partial x_j} \rangle = \frac{\partial \psi}{\partial x_i}\left(\frac{\partial \psi}{\partial x_j}\right) \quad \langle \frac{\partial \psi}{\partial x_j}, \frac{\partial \psi}{\partial x_i} \rangle = \frac{\partial \psi}{\partial x_j}\left(\frac{\partial \psi}{\partial x_i}\right)$$

Thus,

$$\frac{\partial \psi}{\partial x_j}\left(\frac{\partial \psi}{\partial x_i}\right) - \frac{\partial \psi}{\partial x_i}\left(\frac{\partial \psi}{\partial x_j}\right) = 0$$

Rectilinear Coordinates

Any coordinate system, in which the **Orthogonality** of Tangent Space exists, is called **Rectilinear Coordinates**.

$$g_{ij} = \langle \frac{\partial}{\partial x_i}, \frac{\partial}{\partial x_j} \rangle = g_{ji} = \langle \frac{\partial}{\partial x_j}, \frac{\partial}{\partial x_i} \rangle = \delta_{ji}$$

However, not all coordinate systems can satisfy such condition. Therefore, in Rectilinear Coordinates, if ψ be a Vector Field with the basis $\{ \boldsymbol{\partial}_1, \ldots, \boldsymbol{\partial}_n \}$, then we have a diagonal matrix.

$$\langle \frac{\partial \psi}{\partial x_i}, \frac{\partial \psi}{\partial x_j} \rangle = \begin{pmatrix} \frac{\partial^2 \psi}{\partial \boldsymbol{x}_1^2} & \cdots & 0 \\ \vdots & \ddots & \vdots \\ 0 & \cdots & \frac{\partial^2 \psi_n}{\partial \boldsymbol{x}_n^2} \end{pmatrix} = \frac{\partial^2 \psi}{\partial^2 \boldsymbol{x}_k}$$

Curvilinear Coordinates

Any coordinate system, in which the **Orthogonality** of Tangent Space does not exist, is called **Curvilinear Coordinates**.

That's why in Curvilinear Coordinates, we have to define the **Metric Connections**, when we parallel transport a vector from one tangent space to another, so that their **Inner Product Space** remains the same, as much as possible.

As we already know, the Metric Tensor $\mathbf{g}_{ij}$ is represented as:

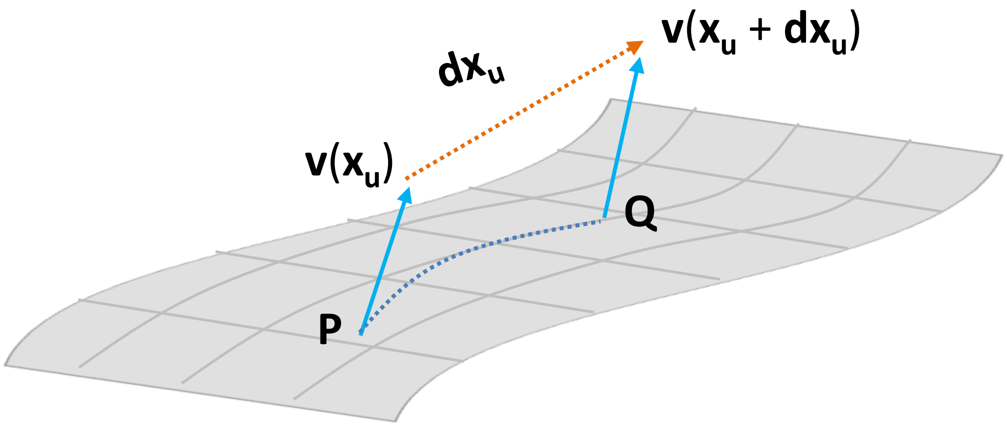
$$\langle \frac{\partial \psi}{\partial x_i}, \frac{\partial \psi}{\partial x_j} \rangle \quad \text{and} \quad \langle \frac{\partial \psi}{\partial x_j}, \frac{\partial \psi}{\partial x_i} \rangle$$

Therefore, there can be different connections, depending on how we define:

$$\frac{\partial^2 \psi}{\partial \boldsymbol{x}_i \partial \boldsymbol{x}_j} \quad \text{and} \quad \frac{\partial^2 \psi}{\partial \boldsymbol{x}_j \partial \boldsymbol{x}_i}$$

Affine Connection

As we already know, each Tangent Space $\mathbf{T_pR^n}$ in a Tangent Bundle **TP** are distinct, and therefore, if we want to compare a vector **v** from a Tangent Space at **P** to a Tangent Space at **Q**, we need to define the **Metric Connection**, or simply a **Connection**, especially in Curve Space.



Since **v** is being transported from point P to point Q, without changing, we assume it is constant, such that:

$$\nabla_{\partial_i} \boldsymbol{v} = \nabla_i \boldsymbol{v} = \mathbf{0}$$

Affine Connection

Since $$\nabla_{\boldsymbol{\partial}_i}\boldsymbol{v} = \nabla_i\boldsymbol{v} = \boldsymbol{0}$$

$$\nabla_{\boldsymbol{\partial}_i}\left(v^j\boldsymbol{\partial}_{\boldsymbol{j}}\right) = \frac{\partial v^j}{\partial \boldsymbol{x}_i}\boldsymbol{\partial}_j + v^j\Gamma_{ij}^k\boldsymbol{\partial}_k = 0$$

Thus, $$\frac{\partial v^j}{\partial \boldsymbol{x}_i} = -v^j\Gamma_{ij}^k$$

$$\lim_{dx_u \to 0}\frac{v^j(x_u + dx_u) - v^j(x_u)}{dx_u} = -v^j\Gamma_{ij}^k$$

$$dv^j \approx -v^j\Gamma_{ij}^k dx_u$$

Therefore, $$v^j(x_u + dx_u) \approx v^j(x_u) - v^j\Gamma_{ij}^k dx_u$$

Affine Connection

As we can see, the **Christoffel Symbols** represents the coefficients of the Affine Connection. In other words, the difference between vector **v** at point P and vector v at point Q is defined by **Christoffel Symbols**.

$$dv^j \approx -v^j \Gamma_{ij}^{k} dx_u$$

$$v^j(x_u + dx_u) \approx v^j(x_u) - v^j \Gamma_{ij}^{k} dx_u$$

However, please be aware that **Christoffel Symbols** are **NOT** Tensors, but they are the coefficients of metric connection, which represent the following:

$$\frac{\partial^2 \psi}{\partial \boldsymbol{x}_i \partial \boldsymbol{x}_j} \text{ and } \frac{\partial^2 \psi}{\partial \boldsymbol{x}_j \partial \boldsymbol{x}_i}$$

Levi-Civita Connection

An **affine connection ∇** is defined as **Levi-Civita Connection** if it has the following properties:

$$\nabla g = 0$$

$$\nabla_{\partial_i}\partial_j - \nabla_{\partial_j}\partial_i = \nabla_i\partial_j - \nabla_j\partial_i = 0$$

Now, let us look more into these properties. The first property suggests that the **Gradient of Metric Tensor ∇g** must be zero, which implies that the **Change of Metric Tensor** must be preserved.

The second property suggests that the **Difference of Christoffel Symbols** $\nabla_i\partial_j$ and $\nabla_j\partial_i$ must be zero, which implies that Torsion must be zero, as the **Difference of Christoffel Symbols** is defined as **Torsion Tensor**.

$$\nabla_i\partial_j - \nabla_j\partial_i = T$$

Levi-Civita Connection

Levi-Civita Connection can be actually defined as:

$$X.g(Y,Z) = X^i \boldsymbol{\partial}_i (g_{jk} Y^j Z^k)$$

$$X.g(Y,Z) = (\nabla_x g)(Y,Z) + g(\nabla_x Y, Z) + g(Y, \nabla_x Z)$$

In other words, **Levi-Civita Connection** is the covariant derivative of the scalar field defined by the Metric Tensor between Y and Z Vector Spaces, with respect to X Vector Space.

However, since the covariant derivative of g is zero,

$$(\nabla_x g)(Y,Z) = 0$$

Then,

$$X.g(Y,Z) = g(\nabla_x Y, Z) + g(Y, \nabla_x Z)$$

Parallel Transport

Now, we can formally define what really is Parallel Transport, though we have briefly learned it before.

Given a curve $\lambda \subseteq R \longrightarrow U \subseteq R^n$ and a vector field v: $U \longrightarrow Tp$, we can say that the vector field v is **parallel transported** along the curve λ if, for every $t \in I$:

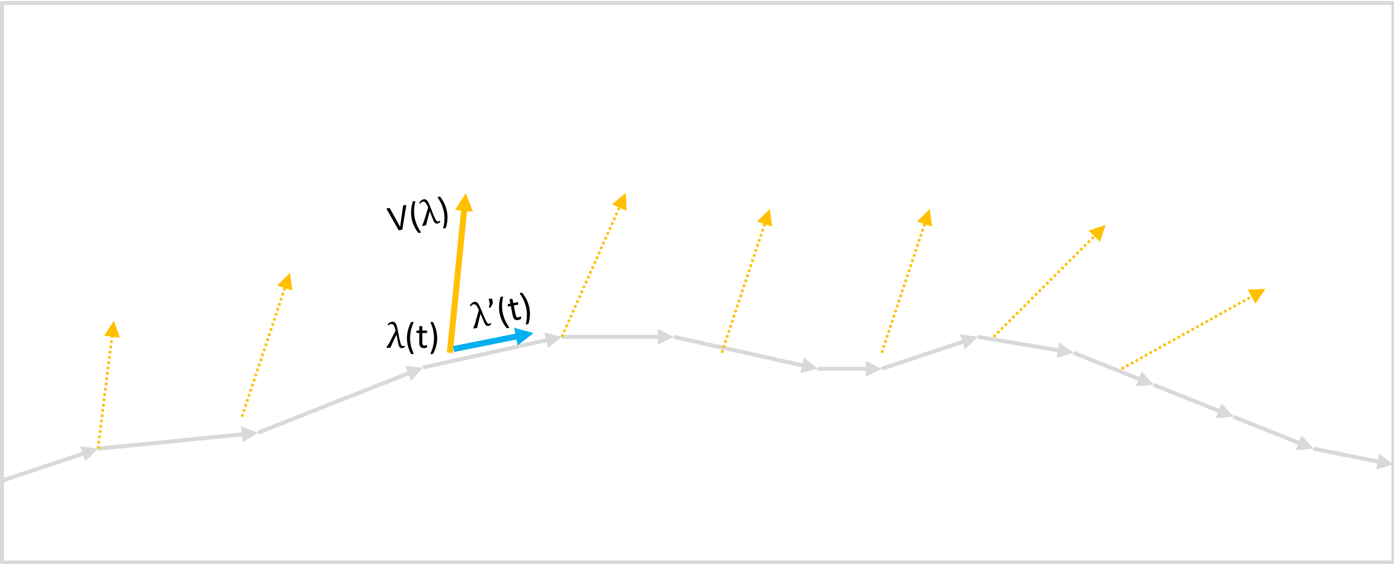
$$\nabla_{\lambda'(t)} V\big(\lambda(t)\big) = 0$$

In other words, if the covariant derivative of a Vector Field **V** with respect to a Velocity of Curve $\boldsymbol{\lambda}$ is zero, its metric tensor is said to be preserved.

Parallel Transport

In order to understand the Parallel Transport, we can visualize Parallel Transport as a Connection **V** being transported along the Curve λ(t).

$$\nabla_{\lambda'(t)} V\big(\lambda(t)\big) = 0$$



Since the Covariant Derivative of V(λ(t)) is zero, its **Metric Connection** must be constant.

Gradient of Metric Tensor

Again, let us define the Gradient of Metric Tensor, and let us look into why it is important.

As we know already, the Metric Tensor is defined as:

$$g_{ij} = \langle \frac{\partial}{\partial x_i}, \frac{\partial}{\partial x_j} \rangle = \frac{\partial}{\partial x_i} \cdot \frac{\partial}{\partial x_j}$$

Thus, the **Gradient of Metric Tensor** with respect to Vector Field can be defined as:

$$\nabla_{\partial_k} g_{ij} = \frac{\partial g_{ij}}{\partial_k} = \frac{\partial \psi}{\partial \boldsymbol{x}_k} \langle \frac{\partial \psi}{\partial \boldsymbol{x}_i}, \frac{\partial \psi}{\partial \boldsymbol{x}_j} \rangle$$

$$= \langle \frac{\partial^2 \psi_1}{\partial \boldsymbol{x}_k \partial \boldsymbol{x}_i}, \frac{\partial \psi}{\partial \boldsymbol{x}_j} \rangle + \langle \frac{\partial \psi}{\partial \boldsymbol{x}_i}, \frac{\partial^2 \psi}{\partial \boldsymbol{x}_k \partial \boldsymbol{x}_j} \rangle$$

Gradient of Metric Tensor

However, we also have to consider all combinations of $\nabla_{\partial_k} g_{ij}, \nabla_{\partial_i} g_{jk}, \nabla_{\partial_j} g_{ki}$.

Thus, the **Gradient of Metric Tensor** with respect to Vector Field can be defined as:

$$\nabla_{\partial_k} g_{ij} = \langle \frac{\partial^2 \psi_1}{\partial \boldsymbol{x}_k \partial \boldsymbol{x}_i}, \frac{\partial \psi}{\partial \boldsymbol{x}_j} \rangle + \langle \frac{\partial \psi}{\partial \boldsymbol{x}_i}, \frac{\partial^2 \psi}{\partial \boldsymbol{x}_j \partial \boldsymbol{x}_k} \rangle$$

$$\nabla_{\partial_i} g_{jk} = \langle \frac{\partial^2 \psi_1}{\partial \boldsymbol{x}_i \partial \boldsymbol{x}_j}, \frac{\partial \psi}{\partial \boldsymbol{x}_k} \rangle + \langle \frac{\partial \psi}{\partial \boldsymbol{x}_j}, \frac{\partial^2 \psi}{\partial \boldsymbol{x}_k \partial \boldsymbol{x}_i} \rangle$$

$$\nabla_{\partial_j} g_{ik} = \langle \frac{\partial^2 \psi_1}{\partial \boldsymbol{x}_i \partial \boldsymbol{x}_j}, \frac{\partial \psi}{\partial \boldsymbol{x}_k} \rangle + \langle \frac{\partial \psi}{\partial \boldsymbol{x}_i}, \frac{\partial^2 \psi}{\partial \boldsymbol{x}_j \partial \boldsymbol{x}_k} \rangle$$

Thus,

$$\nabla_{\partial_i} g_{jk} + \nabla_{\partial_j} g_{ki} - \nabla_{\partial_k} g_{ij} = 2 \langle \frac{\partial \psi}{\partial \boldsymbol{x}_k}, \frac{\partial^2 \psi}{\partial \boldsymbol{x}_i \partial \boldsymbol{x}_j} \rangle$$

Gradient of Metric Tensor

Since

$$\nabla_{\partial_i} g_{jk} + \nabla_{\partial_j} g_{ki} - \nabla_{\partial_k} g_{ij} = 2\left\langle \frac{\partial \psi}{\partial \boldsymbol{x}_k}, \frac{\partial^2 \psi}{\partial \boldsymbol{x}_i \partial \boldsymbol{x}_j} \right\rangle$$

But already know,

$$\left\langle \frac{\partial \psi}{\partial \boldsymbol{x}_l}, \frac{\partial^2 \psi}{\partial \boldsymbol{x}_i \partial \boldsymbol{x}_j} \right\rangle = g_{kl} \Gamma_{ij}^k$$

Thus,

$$\nabla_{\partial_i} g_{jk} + \nabla_{\partial_j} g_{ki} - \nabla_{\partial_k} g_{ij} = 2 g_{kl} \Gamma_{ij}^k$$

$$\Gamma_{ij}^k = \frac{1}{2} g^{kl} \left(\nabla_{\partial_i} g_{jk} + \nabla_{\partial_j} g_{ki} - \nabla_{\partial_k} g_{ij} \right)$$

It is how Christoffel Symbols are related to the Gradient of Metric Tensor. In a sense, Christoffel Symbols can represent how Metric Tensor would vary in Tangent Spaces.

Gradient of Fields

Now, let us formally define **Gradient of Fields** in general. As we already know, the Jacobian Matrix **J** is a differential map, which can map from one vector space to another via linear approximation.

Analogously, we can represent a differential map **J** as a connection **∇**, which can connect different Tangent Spaces.

In this sense, we can see gradient as multi-differential map or connection, similar to multi-linear map.

Obviously, such multi-differential map is a **Tensor**, and thus, gradient of fields are **K-Differential Map** or **K-Differential Connection** or simply **Tensor Fields**.

K-Differential Map

As we already know, the Covariant Derivative is defined as:

$$\nabla_{\boldsymbol{v}} \boldsymbol{W} = \nabla_{v^i \boldsymbol{\partial}_i} (w^j \boldsymbol{\partial}_j)$$

We can assume that Covariant Derivative defines the connection between Vector Fields **V** and **W**. Now, let us consider the connection between a Vector Field W and the basis of vector field ∂_{i}.

$$\nabla_{\boldsymbol{\partial}_i} \boldsymbol{W} = \nabla_i \boldsymbol{W}$$

Then,

$$\nabla \boldsymbol{W} = \frac{\partial \boldsymbol{W}}{\partial \boldsymbol{x}_i} \otimes \boldsymbol{\partial}^i = \frac{\partial \left(w_j \, \boldsymbol{\partial}_j \right)}{\boldsymbol{\partial} x_i} \otimes \boldsymbol{\partial}^i$$

$$= \frac{\partial w_j}{\partial \boldsymbol{x}_i} \, \boldsymbol{\partial}_j \otimes \boldsymbol{\partial}^i$$

K-Differential Map

Now, we can generally define a **K-Differential Map A**(x) ∈ $\otimes^k T_p$ as

$$\nabla \boldsymbol{A}(x) = \sum \partial_j A_{i_1 \cdots i_k}(x)\, \boldsymbol{\partial}_{i_1} \otimes \cdots\ \boldsymbol{\partial}_{i_k} \otimes \boldsymbol{\partial}^j$$

However, as we are aware, we need to take account for **Metric Tensor** for **Rectilinear** Coordinates and **Curvilinear** Coordinates.

K-Differential Map

Now, let us look into a **K-Differential Map A**(x) ∈ $\otimes^k T_p$ in **Rectilinear** Coordinates.

When k = 0, then we get a Gradient or 0-Differential Map of Scalar Field,

$$\nabla \mathrm{f}(\mathrm{x}) = \frac{\partial f_i}{\partial \boldsymbol{x}_i} \; \boldsymbol{\partial}^i$$

When k=1, then we get a 1-Differential Map ,

$$\nabla \mathbf{V}(\mathrm{x}) = \frac{\partial v_j}{\partial \boldsymbol{x}_i} \; \boldsymbol{\partial}_j \otimes \boldsymbol{\partial}^i$$

When k=2, then

$$\nabla \mathbf{W}(\mathrm{x}) = \frac{\partial w_{jk}}{\partial \boldsymbol{x}_i} \; \boldsymbol{\partial}_j \otimes \boldsymbol{\partial}_k \otimes \boldsymbol{\partial}^i$$

K-Differential Map

Again, let us look into a **K-Differential Map A**(x) ∈ $\otimes^k T_p$ in **Curvilinear** Coordinates. However, as we know, we need to add **Christoffel Symbols** as a correction terms.

When k = 0,

$$\nabla \mathrm{f}(\mathrm{x}) = \frac{\partial f_i}{\partial \boldsymbol{x}_i} \; \boldsymbol{\partial}^i$$

When k=1,

$$\nabla \mathbf{V}(\mathrm{x}) = \left(\frac{\partial v_j}{\partial \boldsymbol{x}_i} + v_k \Gamma_{ij}^k \right) \; \boldsymbol{\partial}_j \otimes \boldsymbol{\partial}^i$$

When k=2,

$$\nabla \mathbf{W}(\mathrm{x}) = \left(\frac{\partial w_{jk}}{\partial \boldsymbol{x}_i} + w_{lk} \Gamma_{ij}^l + w_{jl} \Gamma_{ik}^l \right) \; \boldsymbol{\partial}_j \otimes \boldsymbol{\partial}_k \otimes \boldsymbol{\partial}^i$$

Cotangent Space

Previously, we defined the Tangent Space $\mathbf{T_pR^n}$. Now, we will define the Cotangent Space $\mathbf{T^*{}_pR^n}$, which is the Dual Space of Tangent Space $\mathbf{T_pR^n}$.

Let $p \in$ R^n. The cotangent space to R^n at p is the dual vector space, and thus an element of T*$_p$R^n is called a cotangent vector to R^n at p.

$$T^*{}_p\boldsymbol{R}^n \rightarrow \{(\boldsymbol{p},\boldsymbol{v}^*);\ \boldsymbol{v}^*\ \in \boldsymbol{U},\boldsymbol{p}\ \in \boldsymbol{V}\}$$

$$(\boldsymbol{p},\boldsymbol{v}^*) \rightarrow \boldsymbol{v}^*: \boldsymbol{v}^*(\boldsymbol{p})$$

Basis of Cotangent Space

if **v*** be a Cotangent Vector in Cotangent Space $\mathbf{T^*}_{\mathbf{p}}$, then we can define:

$$\omega = \boldsymbol{v}^*(\boldsymbol{x}) = \sum_{i=1}^{n} f_i(\boldsymbol{x})dx_i$$

Therefore, the basis of Cotangent Space $\mathbf{T^*}_{\mathbf{p}}$ is defined as { dx_1, …, dx_n} the dual basis of Tangent Space $\mathbf{T}_{\mathbf{p}}$, the basis of which is {$\partial / \partial x_1$, …, $\partial / \partial x_n$ }.

$$\delta_j^i = \sum_{i=1}^{n} dx_i \frac{\partial}{\partial \boldsymbol{x}_j}$$

Differential Form

Therefore, the Linear Form of Differential Map can be represented by Differential Form.

Differential Form is the Linear Form of Cotangent Vector v* ∈ $T^*{}_p R^n$, and thus, it can be represented in:

$$\omega = \boldsymbol{v}^*(\boldsymbol{x}) = \sum_{i=1}^{n} f_i(\boldsymbol{x}) dx_i$$

More formally, we will use the notation of **d*f*** , which is the Cotangent of **∇*f*,** to represent the differential form such that:

$$df = \sum_{i=1}^{n} \frac{\partial f}{\partial \boldsymbol{x}_i} d\boldsymbol{x}_i$$

Diffeomorphism

Let V ⊂ R^n and W ⊂ R^m be open, and let dφ: V ⟶ W as φ: p ⟶ q be a C$^\infty$ map. If $\boldsymbol{v}$ is a C$^\infty$-vector field on V and $\boldsymbol{w}$ a C$^\infty$-vector field on W, we say that $\boldsymbol{v}$ and $\boldsymbol{w}$ are φ-related if, for all p ∈ V, then:

$$q = \varphi(p)$$

$$d\varphi_p(\boldsymbol{v}(p)) = \boldsymbol{w}(\varphi(p))$$

But,

$$\boldsymbol{v} = \sum_{i=1}^{n} v_i \frac{\partial}{\partial \boldsymbol{x}_i} \qquad \boldsymbol{w} = \sum_{j=1}^{m} w_j \frac{\partial}{\partial \boldsymbol{y}_j}$$

Diffeomorphism

Since,

$$d\varphi_p(\boldsymbol{v}(p)) = \boldsymbol{w}(\varphi(p))$$

We can see,

$$w_j(q) = \sum_{j=1}^{n} \frac{\partial \varphi_i}{\partial \boldsymbol{x}_j}(p) v_j\,(p)$$

$$w_j = \sum_{j=1}^{n} \frac{\partial \varphi_i}{\partial \boldsymbol{x}_j} v_j \circ \varphi^{-1}$$

Thus, vector field $\boldsymbol{w}$ on W has a property that $\boldsymbol{v}$ and $\boldsymbol{w}$ are φ –related.

Pushforward (f_*)

We can denote the vector field $\boldsymbol{w}$ by $f_*\boldsymbol{v}$, and call $f_*\boldsymbol{v}$ the pushforward of $\boldsymbol{v}$ by f, if $\boldsymbol{w}$ is f-related to $\boldsymbol{v}$.

$$q = f(p)$$

$$df_p(\boldsymbol{v}(p)) = \boldsymbol{w}(f(p))$$

$$df_p(\boldsymbol{v}) = \boldsymbol{w} \circ f$$

$$df_p(\boldsymbol{v}) \circ f^{-1} = \boldsymbol{w}$$

$$df_p \circ f^{-1}(\boldsymbol{v}) = \boldsymbol{w}$$

$$f_*\boldsymbol{v} = \boldsymbol{w}$$

Pullback (f^*)

if $\boldsymbol{w}$ is f-related to $\boldsymbol{v}$, given a one-form μ on **W**, we can define a pullback one-form $f^*\mu$ on **V** by the following:

$$\mu : W \to R \quad q = f(p)$$

And,

$$df_p(\boldsymbol{v}(p)) = \boldsymbol{w}(f(p))$$

Then,

$$\mu_{f(p)} \circ df_p(\boldsymbol{v}) \to R \in T^*{}_p\boldsymbol{R}^n$$

$$\mu_{f(p)} \circ df_p(\boldsymbol{v}) = \mu_q(\boldsymbol{w}(q))$$

$$\mu_q \circ df_p = d\varphi_q \circ df_p = d(\varphi \circ f)_p$$

$$if\ \mu = d\varphi, \qquad f^*\mu = d\varphi \circ f$$

$$f^*w = v^*$$

Pushforward and Pullback

Let **W** and V be open subsets of R^n and f: **V** $\longrightarrow$ **W** is a diffeomorphism. If **v** is a vector field on V, we can define the pullback of **v** to **V**, to be the vector field:

$$df_p : T_p R^n \to T_{f(p)} R^n$$

$$df^*{}_p : T^*{}_{f(p)} R^n \to T^*{}_p R^n$$

Thus,

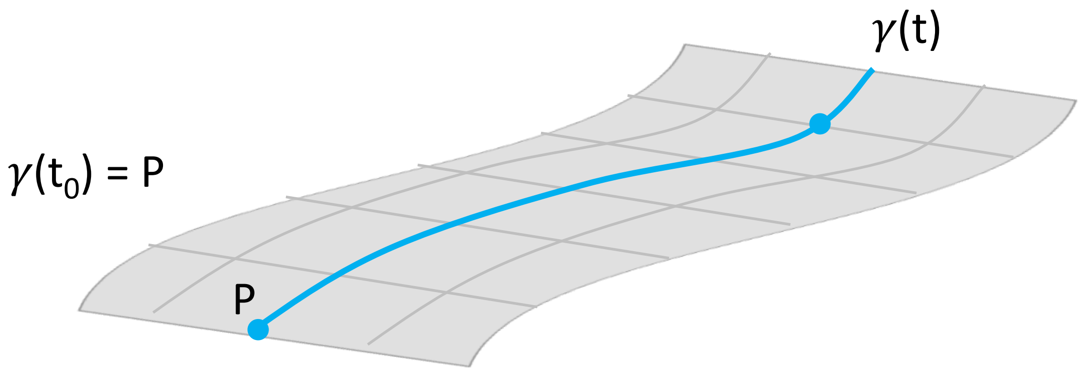
$$f_* v = w$$

$$f^* w = v^*$$

$$f^* w = (f_*^{-1} w)$$

Integral Curve

Let $U \subset \mathbf{R}^n$ be open and $\boldsymbol{v}$ a vector field on U. A C^1 curve γ: $(a,b) \longrightarrow U$ is an integral curve of $\boldsymbol{v}$ if for all $t \in (a,b)$, then:



$$\boldsymbol{v}\big(\gamma(t)\big) = (\gamma(t), \frac{d\gamma}{dt}(t))$$

If $\boldsymbol{v}$ is the vector field, and $\boldsymbol{g}$: $U \to \mathbf{R}^n$ is the function $(\boldsymbol{g}_1, \ldots, \boldsymbol{g}_n)$, the condition for $\gamma(t)$ to be an integral curve of $\boldsymbol{v}$ is that it satisfy the system of differential equations:

$$\frac{d\gamma}{dt}(t) = g\big(\gamma(t)\big) = \boldsymbol{v}_p(\gamma(t))$$

Properties of Integral Curve

Let $U \subset \mathbf{R}^n$ be open and $\boldsymbol{v}$ a vector field on U. A C^1 curve γ: $(a,b) \longrightarrow U$ is an integral curve of $\boldsymbol{v}$ if for all $t \in (a,b)$, then the γ(t) can have the following properties:

$$\gamma(t_0) = p$$

$$\frac{d\gamma}{dt}(t_0) = \boldsymbol{v}_p$$

If C^1 function ϕ: $U \longrightarrow \mathbf{R}$ is an **integral** of the systems of equations if for every integral curve $\gamma(t)$, the function $t \mapsto \phi(\gamma(t))$ is constant.

$$\frac{d}{dt}\varphi\big(\gamma(t)\big) = D(\varphi)_{\gamma(t)}(\boldsymbol{v}) = 0$$

Flow of Vector Field

If $\boldsymbol{v}$ is **complete** for every p, then the flow of vector field $\boldsymbol{v}$ is the one parameter group of diffeomorphisms such that

$$\varphi_t : U \to U$$

For each point p ∈ U, the flow line (trajectory) of p is given by ϕ_t(p), and the flow of $\boldsymbol{v}$ is represented by the systems of equations:

$$\frac{d}{dt}\varphi_t(p) = \boldsymbol{v}(\varphi_t(p))$$

The flow of vector field $\boldsymbol{v}$ has the following properties.

$$\varphi_0(p) = p$$

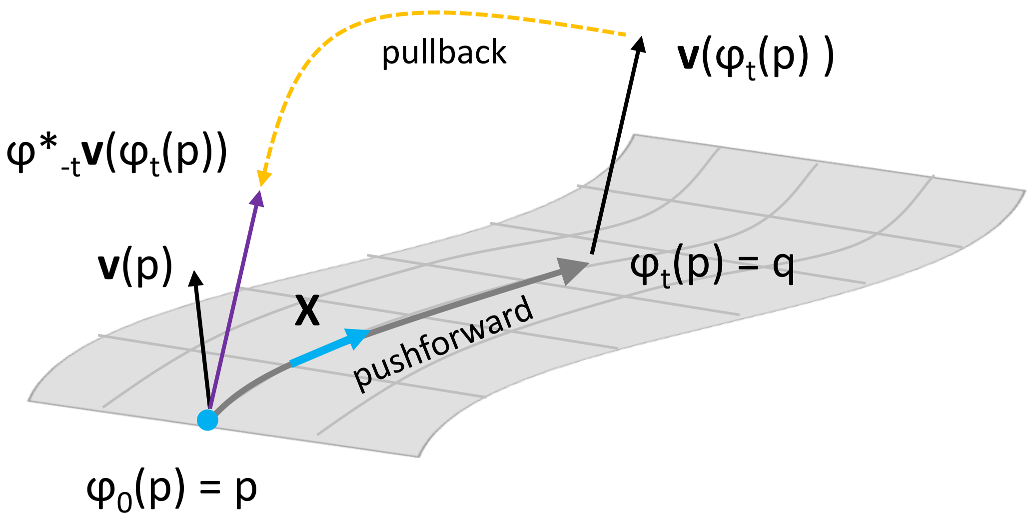
$$\frac{d}{dt}\varphi_t(p)\Big|_{t=0} = v_p$$

$$\varphi_{t+s} = \varphi_t \circ \varphi_s$$

$$\varphi_{-t} = {\varphi_t}^{-1}$$

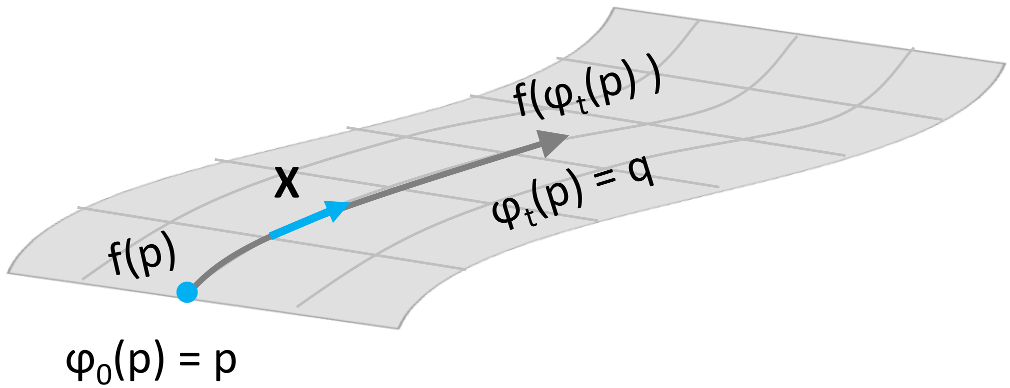
Flow of Vector Fields

With the definition of Vector Flow, we can actually represent **pushforward** and **pullback** operations as **transporting** (**dragging**) a vector **v** along the flow line (trajectory) of vector field **X**.



Lie Derivative

Let dφ: X ⟶ X as φ: p ⟶ q be a C^{∞} map, then function *f*: φ(p) ⟶ φ(q).



Then, $L_x f$ *, which is known as* Lie Derivative, pronounced as "*Lee Derivative",* named after *"**Sophus Lie**",* can be defined as:

$$L_x f = \lim_{t \to 0} \frac{\varphi_t{}^* f\big(\varphi_t(p)\big) - f(\varphi_0(p))}{t}$$

Lie Derivative

Since

$$L_x f = \lim_{t \to 0} \frac{\varphi_t{}^* f\big(\varphi_t(p)\big) - f(\varphi_0(p))}{t}$$

$$\varphi_t{}^* f\big(\varphi_t(p)\big) = f\big(\varphi_t(p)\big) \qquad = \frac{\partial}{\partial t} f(\varphi_0(p))$$

$$= \frac{\partial f}{\partial \varphi_0(p)} \frac{\partial \varphi_0(p)}{\partial t}$$

$$= \frac{\partial x^i(t)}{\partial t} \frac{\partial f}{\partial x^i(t)}$$

$$= X \frac{\partial f}{\partial x^i}$$

$$= X(f)$$

Lie Derivative

We can define Lie Derivative of Vector Field **Y** with respect to Vector Field **X** such that

$$L_X Y(f) = X\big(Y(f)\big) - Y(X(f))$$

And [X, Y] is defined as **Lie Bracket**, which we will look into later,

$$X\big(Y(f)\big) - Y\big(X(f)\big) = [X, Y](f)$$

Differential K-Forms

As we know, Linear Form (1-Form) of Cotangent Vector $v^* \in T^*{}_p R^n$ can be represented with Exterior Product or Wedge Product.

$$\Lambda^1\left(T^*{}_p\boldsymbol{R}^n\right) = T^*{}_p\boldsymbol{R}^n$$

Therefore, if there is k^th Exterior Product, we can have:

$$\Lambda^k\left(T^*{}_p\boldsymbol{R}^n\right)$$

Thus, Differential K-Forms can be defined with k^th Exterior Product of Cotangent Vectors. If U be an open subset of R^n, then k-form ω on U is a function which assigns to each point $p \in U$ an element $\omega_p \in \Lambda^k(T^*{}_p\mathrm{R}^n)$.

Differential K-Forms

Let $\omega_1,\ldots,\omega_k$ be one-forms. Then $\omega_1 \wedge \cdots \wedge \omega_k$ is the k-form whose value at p is the wedge product:

$$(\omega_1 \wedge \cdots \wedge \omega_n)_p = (\omega_1)_p \wedge \cdots \wedge (\omega_n)_p$$

If ω = df, then

$$\omega_1 \wedge \cdots \wedge \omega_n = df_1 \wedge \cdots \wedge df_n$$

Unless otherwise stated, that all the k-forms are of class C^∞. We denote the set of k-forms of class C^∞ on U by $\Omega^k(U)$.

Basis of Differential K-Forms

Since the basis of Cotangent Space $\mathbf{T^*_p}$ is defined as $\{ dx_1, \ldots, dx_n\}$ the dual basis of Tangent Space $\mathbf{T_p}$, then the basis of $\Lambda^k(T^*{}_p\mathrm{R}^n)$ is:

$$dx_{I_p} = (dx_1)_p \wedge \cdots \wedge (dx_n)_p$$

Then, every element of $\Lambda^k(T^*{}_p\mathrm{R}^n)$ can be written uniquely as a sum:

$$\omega = \sum f_I dx_I$$

Exterior Derivative

Let U be an open subset of R^n, we can define an **Exterior Differentiation** as:

$$d\colon \Omega^k(U) \to \Omega^{k+1}(U)$$

If $\omega_1 \in \Omega^k(U)$ and $\omega_1 \in \Omega^l(U)$, then the exterior derivative has the following properties:

$$d(\omega_1 + \omega_2) = d\omega_1 + d\omega_2$$

$$d(\omega_1 \wedge \omega_2) = d\omega_1 \wedge \omega_2 + (-1)^k \omega_1 \wedge d\omega_2$$

$$d(d\omega) = 0$$

Exterior Derivative

Since,

$$\omega = \sum f_I dx_I$$

$$\omega_1 \wedge \omega_2 = \sum d(f_I g_I)\, dx_I \wedge dx_J$$

Then,

$$d\omega = \sum df_I \wedge dx_I$$

$$d(\omega_1 \wedge \omega_2) = \sum d(f_I g_I) \wedge dx_I \wedge dx_J$$

Exterior Derivative

Since,

$$d(d\omega) = \sum d(df_I) \wedge dx_I = 0$$

Then,

$$d(df) = \sum_{j=1}^{n} d(\frac{\partial f}{\partial \boldsymbol{x}_j})\, dx_j \; = \sum_{j=1}^{n} \left(\sum_{i=1}^{n} \frac{\partial^2 f}{\partial \boldsymbol{x}_i \partial \boldsymbol{x}_j}\, dx_i \right) \wedge dx_j$$

$$= \sum_{i,j=1}^{n} \frac{\partial^2 f}{\partial \boldsymbol{x}_i \partial \boldsymbol{x}_j}\, dx_i \wedge dx_j$$

Since, $dx_i \wedge dx_j = -dx_j \wedge dx_i$

$$\frac{\partial^2 f}{\partial \boldsymbol{x}_j \partial \boldsymbol{x}_i} = \frac{\partial^2 f}{\partial \boldsymbol{x}_i \partial \boldsymbol{x}_j}$$

Exact and Closed K-Form

Let U be an open subset of R^n. Then k-form $\omega \in \Omega^k$(U) is **closed** if $d\omega$ = 0, and k-form is **exact** if ω = $d\mu$ for some $\mu \in \Omega^{k-1}$(U).

Every exact form is closed; however, the converse is not true.

Interior Product of Differential Form

Suppose that **v** is a vector field and ω a one-form on $U \subset$ R^n. Now, for every $p \in U$ the vectors **v**$(p) \in \boldsymbol{T_p}$**R**$^{\mathbf{n}}$ and $\omega_p \in \Lambda^k(T^*{}_pR^n)$, then the interior product of $\boldsymbol{v}$ with ω can be defined as the $(k-1)$ Form $\iota_{\boldsymbol{v}}\omega$ on U, which is the element of $\Lambda^{k\text{-}1}(T^*{}_pR^n)$.

$$\iota_{v(p)}(\omega_p)$$

As we can see, it is similar to Covariant Derivative of Vector Field **w** with **v**. However, the major difference is that this interior product is between Tangent Vector **v** and Cotangent Vector or Differential Form ω, instead of the interior product of Tangent Vector **w** with Tangent Vector **v**.

Interior Product of Differential Form

if ω is decomposable, i.e., is a wedge product of one-forms, such that

$$\omega = \mu_1 \wedge \cdots \wedge \mu_k$$

Then,

$$\iota_v(\omega) = \sum_{r=1}^{k} (-1)^{r-1} \iota_v(\mu_r)\ \mu_1 \wedge \cdots \wedge \mu_r \wedge \cdots \wedge \mu_k$$

If $\boldsymbol{v} = \partial/\partial x_i$ and $\omega = dx_I = dx_{i1} \wedge \cdots \wedge dx_{ik}$ then,

$$\iota_v(\omega) = \sum_{i=1}^{k} (-1)^{i-1} \delta_{i,i_r} dx_{I_r}$$

Interior Product of Differential Form

Interior Product of Differential Form have the following properties:

$$\iota_v(\omega_1 + \ \omega_2) = \iota_v(\omega_1) + \iota_v(\omega_2)$$

$$\iota_v(\omega \wedge \mu) = \iota_v\omega \wedge \mu + (-1)^k \omega \wedge \iota_v\mu$$

$$\iota_{v+u}(\omega) = \iota_v(\omega) + \iota_u(\omega)$$

$$\iota_w(\iota_v(\omega)) = -\iota_v(\iota_w(\omega))$$

Lie Derivative

Let U be an open subset of R^n, $\boldsymbol{v}$ a vector field on U, and $\omega \in \Omega^k$(U). The Lie Derivative, pronounced as "*Lee Derivative*", named after "***Sophus Lie***", of ω with respect to $\boldsymbol{v}$ is the k-form, represented by $L_v(\omega)$:

$$L_v(\omega) := \iota_v(d\omega) + d(\iota_v\omega)$$

As we know,

$$\omega = \sum f_I\, dx_I \quad \boldsymbol{v} = \sum g_i \frac{\partial}{\partial \boldsymbol{x}_i}$$

Therefore,

$$L_v(f_I dx_I) = (L_v f_I)dx_I + f_I(L_v dx_I)$$

Lie Derivative

Since

$$L_v(f_I dx_I) = \underline{(L_v f_I)} dx_I + f_I(\underline{L_v dx_I})$$

But,

$$L_v dx_I = \sum_{r=1}^{k} dx_{i_1} \wedge \cdots \wedge L_v dx_{i_r} \wedge \cdots \wedge dx_{i_1} = L_v dx_{i_r}$$

Again,

$$L_v dx_{i_r} = d\, L_v x_{i_r} = g_{i_r}$$

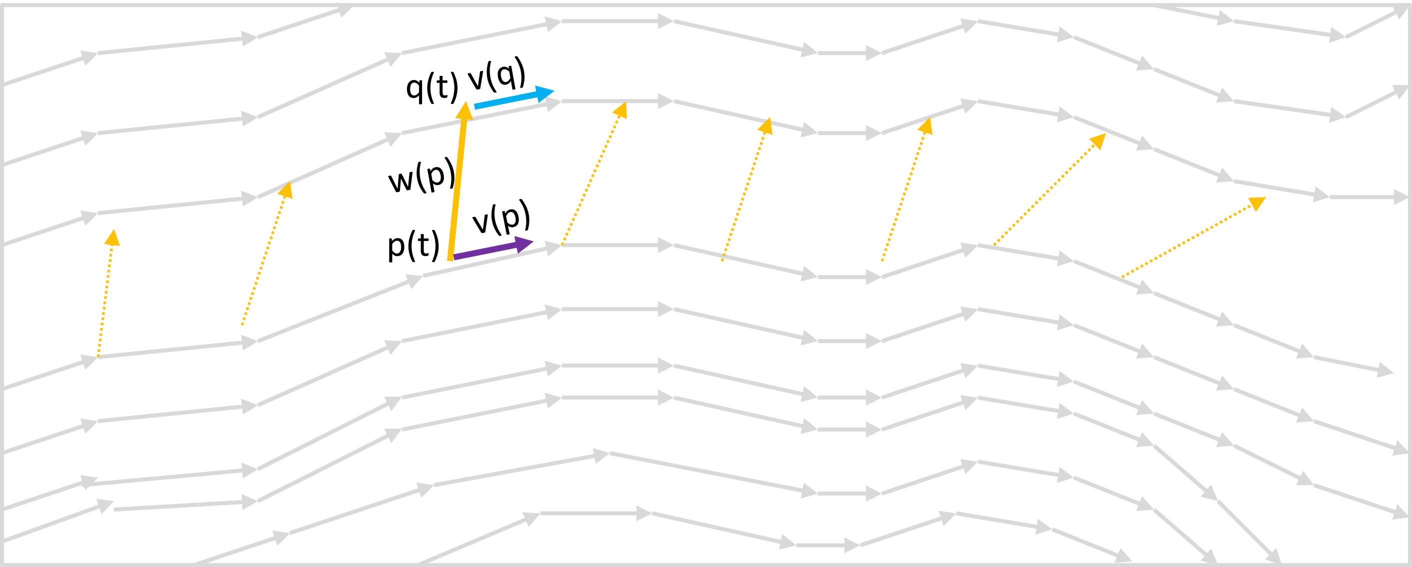
$$L_v f_I = \sum_{i=1}^{n} g_i \frac{\partial f_I}{\partial \boldsymbol{x}_i}$$

Lie Transport

At this point, we will look into **Lie Transport**, which is the underlying principle behind **Lie Derivative**. As we already know **Parallel Transport**, we will define Lie Transport, which will transport a Connection **W**, along the flow of **V**.

$$\lambda \boldsymbol{w} = \boldsymbol{q}(t) - \boldsymbol{p}(t)$$

$$\boldsymbol{p}(t) + \lambda \boldsymbol{w}(\boldsymbol{p}(t)) = \boldsymbol{q}(t)$$



$$By\ \frac{d}{dt},\ \boldsymbol{v}(\boldsymbol{p}) + \lambda \boldsymbol{v} \cdot \nabla \mathbf{w}(\boldsymbol{p}) = \boldsymbol{v}(\boldsymbol{q})$$

Lie Bracket

Since $$\boldsymbol{v}(\boldsymbol{p}) + \lambda \boldsymbol{v} \cdot \nabla \mathbf{w}(\boldsymbol{p}) = \boldsymbol{v}(\boldsymbol{q})$$

$$= \boldsymbol{v}(\boldsymbol{p} + \lambda \boldsymbol{w}(\boldsymbol{p}))$$

If **v** $\approx O(\lambda)$, by **Taylor Expansion**:

$$= \boldsymbol{v}(\boldsymbol{p}) + \lambda \boldsymbol{w} \cdot \nabla \mathbf{v}(\boldsymbol{p}) + O(\lambda^2)$$

Then,

$$\boldsymbol{v}(\boldsymbol{p}) + \lambda \boldsymbol{v} \cdot \nabla \mathbf{w}(\boldsymbol{p}) - \boldsymbol{v}(\boldsymbol{p}) - \lambda \boldsymbol{w} \cdot \nabla \mathbf{v}(\boldsymbol{p}) = O(\lambda^2)$$

If $O(\lambda^2) = 0$,

$$v(\boldsymbol{p}) + \lambda \boldsymbol{v} \cdot \nabla \mathbf{w}(\boldsymbol{p}) = \boldsymbol{v}(\boldsymbol{p}) + \lambda \boldsymbol{w} \cdot \nabla \mathbf{v}(\boldsymbol{p})$$

Then, **Lie Bracket** is defined as:

$$L_v \boldsymbol{w} = [v, \boldsymbol{w}] = \boldsymbol{v} \cdot \nabla \mathbf{w}(\boldsymbol{p}) - \boldsymbol{w} \cdot \nabla \mathbf{v}(\boldsymbol{p}) = 0$$

Lie Bracket

As we can see **Lie Derivative of Vectors** can be defined as **Lie Bracket**:

$$L_v \boldsymbol{w} = [\boldsymbol{v}, \boldsymbol{w}] = \boldsymbol{v} \cdot \nabla \mathbf{w} - \boldsymbol{w} \cdot \nabla \mathbf{v} = 0$$

However, let us redefine it with Vector and Covector Components:

$$L_{\boldsymbol{v}} \boldsymbol{w} = L_{\boldsymbol{v}}\left(w^j \boldsymbol{\partial}_{\boldsymbol{j}}\right) = \underline{L_{\boldsymbol{v}}\left(w^j\right)} \boldsymbol{\partial}_{\boldsymbol{j}} + w^j \underline{L_{\boldsymbol{v}}\left(\boldsymbol{\partial}_{\boldsymbol{j}}\right)}$$

But, for the vector (basis) components,

$$L_{\boldsymbol{v}} \boldsymbol{\partial}_{\boldsymbol{j}} = L_{v^i \boldsymbol{\partial}_{\boldsymbol{i}}}\left(\boldsymbol{\partial}_{\boldsymbol{j}}\right) = v^i \nabla_{\boldsymbol{\partial}_{\boldsymbol{i}}}\left(\boldsymbol{\partial}_{\boldsymbol{j}}\right) - \boldsymbol{\partial}_{\boldsymbol{j}} \nabla_{\boldsymbol{\partial}_{\boldsymbol{i}}}\left(v^i\right)$$

Again, for covector (dual) components,

$$L_{\boldsymbol{v}}\left(w^j\right) = L_{v^i \boldsymbol{\partial}_{\boldsymbol{i}}}\left(w^j\right) = v^i \nabla_{\boldsymbol{\partial}_{\boldsymbol{i}}}\left(w^j\right) + w^j \nabla_{\boldsymbol{\partial}_{\boldsymbol{i}}}\left(v^i\right)$$

Lie Bracket

We will look into why Lie Bracket of Vector and Covector Components are different, by setting **Lie Derivative** and **Covariant Derivative** *equal*.

$$\underline{L_{\boldsymbol{v}}\boldsymbol{w}} = L_{\boldsymbol{v}}\left(w^j\right)\boldsymbol{\partial}_{\boldsymbol{j}} + w^j L_{\boldsymbol{v}}\left(\boldsymbol{\partial}_{\boldsymbol{j}}\right) = \underline{v^i \nabla_{\partial_i}(w^j \boldsymbol{\partial}_{\boldsymbol{j}})}$$

Therefore,

$$v^i \nabla_{\partial_i}(w^j \boldsymbol{\partial}_{\boldsymbol{j}}) = L_{\boldsymbol{v}}\left(w^j\right)\boldsymbol{\partial}_{\boldsymbol{j}} + w^j\left(v^i \nabla_{\partial_i}\boldsymbol{\partial}_{\boldsymbol{j}} - \boldsymbol{\partial}_{\boldsymbol{j}}\nabla_{\partial_i}v^i\right)$$

$$(v^i \nabla_{\partial_i} w^j)\boldsymbol{\partial}_{\boldsymbol{j}} + \underline{w^j v^i \nabla_{\partial_i}\boldsymbol{\partial}_{\boldsymbol{j}}} = L_{\boldsymbol{v}}\left(w^j\right)\boldsymbol{\partial}_{\boldsymbol{j}} + w^j\left(v^i \nabla_{\partial_i}\boldsymbol{\partial}_{\boldsymbol{j}} - \boldsymbol{\partial}_{\boldsymbol{j}}\nabla_{\partial_i}v^i\right)$$

$$= L_{\boldsymbol{v}}\left(w^j\right)\boldsymbol{\partial}_{\boldsymbol{j}} + \underline{w^j v^i \nabla_{\partial_i}\boldsymbol{\partial}_{\boldsymbol{j}}} - w^j \boldsymbol{\partial}_{\boldsymbol{j}}\nabla_{\partial_i}v^i$$

Thus,

$$L_{\boldsymbol{v}}\left(w^j\right)\boldsymbol{\partial}_{\boldsymbol{j}} = (v^i \nabla_{\partial_i} w^j)\boldsymbol{\partial}_{\boldsymbol{j}} + \boldsymbol{\partial}_{\boldsymbol{j}} w^j \nabla_{\partial_i} v^i$$

$$L_{\boldsymbol{v}}\left(w^j\right) = v^i \nabla_{\partial_i} w^j + w^j \nabla_{\partial_i} v^i$$

Deference between Covariant and Lie Derivative

At this point, it should naturally occur to us what is the difference between Covariant Derivative and Lie Derivative.

The **Lie Derivative** (L_v**X**) measures how a tensor field changes as it is transported along the flow of **a vector field** (related to **Symmetries** and **Diffeomorphisms**, requiring no Metric Connection).

$$L_v \boldsymbol{T} = 0$$

The **Covariant Derivative** (∇_v**X**) measures how a tensor field changes when it is parallel transported along **a vector curve**, which requires a choice of connection (like a Metric Connection, such as **Levi-Civita Connection**), defining how to compare vectors at different points.

$$\nabla_v \boldsymbol{T} = 0$$

If both L_v**X** and ∇_v**X** are zero, they are equal, meaning Transport is invariant, such that:

$$L_v \boldsymbol{T} = \nabla_v \boldsymbol{T} = 0$$

Torsion

So far, we have naturally assumed that **Torsion Tensor** is **Zero**, which implies that:

$$L_v \boldsymbol{T} = \nabla_v \boldsymbol{T} = 0$$

Such condition is satisfied by the **Difference of Christoffel Symbols** being equal to zero:

$$\nabla_{\partial_i} \partial_j - \nabla_{\partial_j} \partial_i = T = 0$$

However, Torsion Tensor cannot always be zero. In such cases, we need to look into the **Christoffel Symbols**.

When the Torsion Tensor is not zero, the distortion of Tensor Fields can be ***partially*** represented with two specifics representation: **Divergence** and **Curl**.

Torsion

As we already know, the **metric tensor** is defined such that:

$$g_{ij} = \langle \frac{\partial}{\partial x_i}, \frac{\partial}{\partial x_j} \rangle = \frac{\partial}{\partial x_i} \cdot \frac{\partial}{\partial x_j}$$

And **Christoffel Symbols** represent the **non-diagonal** entries of metric tensor:

$$\frac{\partial^2 \psi}{\partial \boldsymbol{x}_i \partial \boldsymbol{x}_j} \text{ and } \frac{\partial^2 \psi}{\partial \boldsymbol{x}_j \partial \boldsymbol{x}_i}$$

In addition, **Christoffel Symbols** are related to metric tensor such that:

$$\Gamma_{ij}^{k} = \frac{1}{2} g^{kl} \left(\nabla_{\partial_i} g_{jk} + \nabla_{\partial_j} g_{ki} - \nabla_{\partial_k} g_{ij} \right)$$

Such **Christoffel Symbols** are known as **Christoffel Symbols** of **Second Kind**. We can derive **Christoffel Symbols** of **First Kind**.

$$g_{kl} \Gamma_{ij}^{k} = \Gamma_{ijl}$$

Torsion

We can also represent the derivative of metric tensor such that:

$$\partial_\gamma g = g g^{ij} \partial_\gamma g_{ij}$$

From this, we can get:

$$\partial_j g = 2g\Gamma_{ij}^{i}$$

Then, we can get:

$$\Gamma_{ij}^{i} = \frac{1}{2} g^{-1} \partial_j g = \frac{1}{\sqrt{|g|}} \partial_j \sqrt{|g|}$$

Divergence

Now, let φ: $R^n \longrightarrow R$ be a scalar field, then the gradient of φ, which is a vector field, can be defined as:

$$\boldsymbol{v} = \nabla\varphi = \partial_i(\varphi) = \partial_i v^i = v^i\,\partial_i$$

Then, **divergence** of the derivative of scalar field can be defined as:

$$\nabla \cdot \boldsymbol{v} = \nabla \cdot \nabla\varphi$$

As we can see:

$$\begin{aligned}\nabla \cdot \boldsymbol{v} = \nabla_i \cdot \left(v^i\,\partial_i\right) &= \nabla_i v^i\,\partial_i + v^i \nabla_i\,\partial_i \\ &= \partial_i v^i + \Gamma^i_{ij}\,v^j \\ &= \partial_i v^i + \frac{1}{\sqrt{|g|}}\partial_i \sqrt{|g|}\,v^i \\ &= \frac{1}{\sqrt{|g|}}\partial_i\left(\sqrt{|g|}\,v^i\right)\end{aligned}$$

Curl

Now, **Curl** of the derivative of scalar field can be defined as :

$$\nabla \times \boldsymbol{v} = \nabla \times \nabla\varphi$$

If we have the basis of vector field as { $\mathbf{e}_1$, …, $\mathbf{e}_n$ }, then

$$\nabla \times \boldsymbol{v} = e^i\partial_i \times v^j e_j = (e^i \times e_j)\partial_i v^j$$

Then, **Levi-Civita Symbols**, which represents the sign (**orientation**), can be defined as:

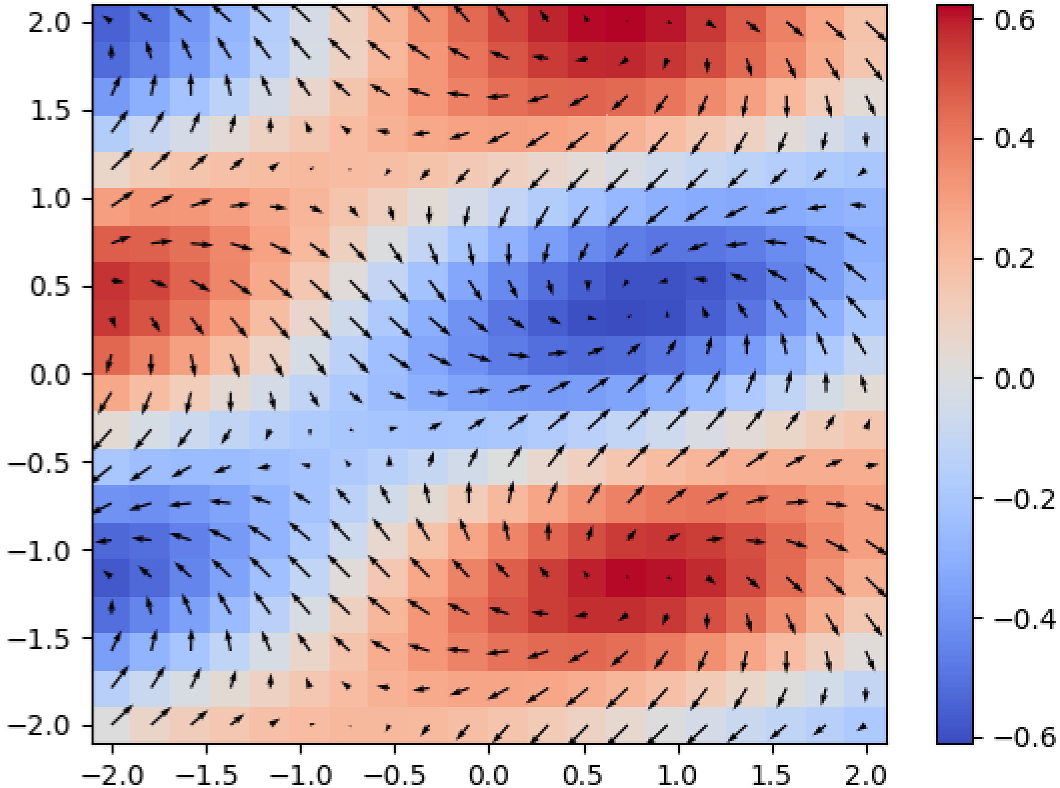
$$\varepsilon^{\delta i}{}_j = (e^i \times e_j)$$

$$\varepsilon^{ijk} = \varepsilon_{ijk} = \begin{cases} -1, & \text{even permutation} \\ 1, & \text{odd permutation} \\ 0, & \text{equal permutation} \end{cases}$$

Non-Zero Torsion

So, when the vector field is **NOT** Torsion Free, then we will have **Curl** and **Divergence**:

$$\nabla \cdot \boldsymbol{v} = \nabla \cdot \nabla \varphi$$



$$\nabla \times \boldsymbol{v} = \nabla \times \nabla \varphi$$

Topology

At this point, we will look into another type of **Mathematical Structure**, known as **Topology**.

Topology is a Mathematical Structure which is preserved under continuous deformations, such as stretching, twisting, crumpling, and bending.

Such deformations are represented in terms of **homeomorphism** and **homotopy**.

Generally, a topology defines a mathematical structure around a neighborhood.

Open and Closed Ball

Suppose V is in Euclidean Space, then

$$B_r(p) = \{ x \in V; ||x - p|| < r \}$$

$$D_r(p) = \{ x \in V; ||x - p|| < r \}$$

$$S_r(p) = \{ x \in V; ||x - p|| < r \}$$

stand for an **open ball**, a **closed ball** and a **sphere** of radius r centered at a point p ∈ V, each of which defines a region around the neighborhood of p.

Now, we can define a neighborhood with Open and Closed Set.

Open and Closed Set

A set $U \subset V$ is called **open** if for any $x \in U$ there exists $r > 0$ such that $B_r(x) \subset U$. And a set $W \subset V$ is called **closed** if its complement $V \setminus W$ is open. As we can see, every open-ball is an open set.

Points which appear as limits of sequences of points $x_n \in W$ are called **limit points** of W.

There are only two subsets of V which are **simultaneously open and closed**: V and Ø.

By a neighborhood of a point $a \in V$, it can be understood as any open set, $p \in U$.

Given any subset $X \subset V$ a point $a \in V$ is called **interior point** for W if there is a neighborhood $p \in U$ such that $U \subset W$, and **boundary point** if it is not an interior point of neither for W nor for its complement $V \setminus W$.

A boundary point of W may or may not belong to W.

The union of all limit points of A is called the **closure** of A and denoted $\bar{A}$.

Compactness and Connectedness

A set $W \subset V$ is called **compact** if W is **closed** and **bounded** with a **boundary**.

A set W is called **path-connected** if for any two points $a_0, a_1 \in W$ there is a continuous path $\gamma : [0,1] \to W$ such that $\gamma(0) = a_0$ and $\gamma(1) = a_1$.

A set W is called **connected** if one cannot present W as a union $W = W_1 \cup W_2$ such that $W_1 \cap W_2 = \emptyset$, $W_1, W_2 = \emptyset$ and both W_1 and W_2 are simultaneously relatively closed and open in W.

Any path-connected set is connected, and any open connected subset $U \subset R_n$ is path connected.

Continuous Map

Let $f : W \longrightarrow V$ be a continuous map, then

1. if W is compact then $f(W)$ is compact;

2. if W is connected then $f(W)$ is connected;

3. if W is path connected then $f(W)$ is path-connected.

As we can see, a topology can represent a continuous map which is connected from one neighborhood from another, preserving its open or closed structure.

Topological Space

A topological space is a pair, (X, τ), consisting of **an underlying set** and **a topology**.

The underlying set is commonly referred to as the **topological space**, and the topology must be a set of subsets of the topological space which is **closed** under arbitrary unions and finite intersections.

A topological isomorphism should take one space onto another and preserve openness.

Homeomorphism

A topological isomorphism is called a homeomorphism.

A homeomorphism between a topological space, (X, τ_X), and a topological space, (Y, τ_Y), is a continuous bijection (a continuous one-to-one onto map with a continuous inverse) from X to Y.

Homotopy

A homotopy between two continuous functions *f* and *g* from a topological space (X, τ_X) to a topological space (Y, τ_Y) is defined as a continuous function $H : X \times [\,0, 1\,] \longrightarrow Y$ from the product of the space *X* with the unit interval [0, 1] to *Y* such that $H(x, 0) = f(x)$, and $H(x, 1) = g(x)$ for all $x \in X$.

From another perspective, a homotopy between two continuous functions $f, g : X \longrightarrow Y$ is a family of continuous functions $h_t : X \longrightarrow Y$ for $t \in [\,0, 1\,]$ such that $h_0 = f$ and $h_1 = g$, and the map $(x, t) \mapsto h_t(x)$ is continuous from $X \times [\,0, 1\,]$ to Y.

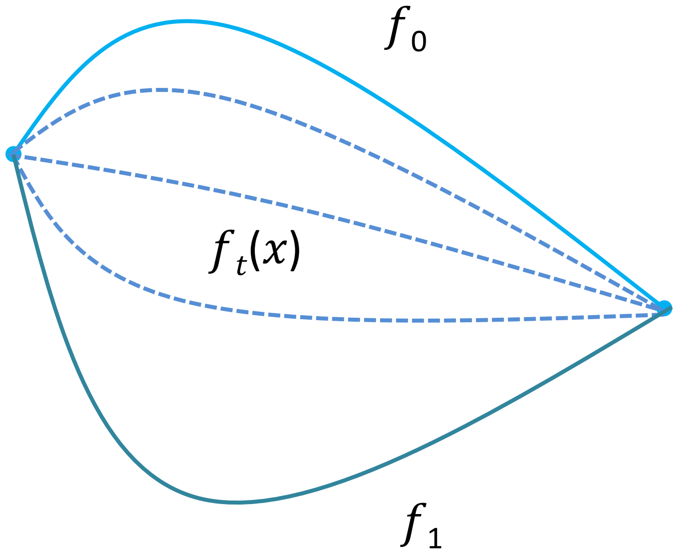
The two versions coincide by setting $h_t(x) = H(x, t)$.

Contractible Space

An open subset U of $\mathbf{R}^n$ is **contractible** if, for some point $p_0 \in U$, then, the identity map id_U: $U \longrightarrow U$ is homotopic to the constant map.

$$f_0 : U \to U, f_0(P) = P_0$$

If there exists a **homotopy** between f_0 and f_1, we can say that f_0 and f_1 are homotopic and write $f_0 \simeq f_1$. Intuitively, f_0 and f_1 are homotopic if there exists a family of C^∞ maps f_t: $U \longrightarrow V$, where $f_t(x) = F(x,t)$, which "**smoothly deform** f_0 into f_1".



Hausdorff Space

A topological space X is **Hausdorff Space** if any pair of distinct points has a corresponding pair of disjoint neighborhoods. That is, $(\forall\ x, y \mid x, y \in X \Rightarrow (\exists G,H \mid G,H \in \tau X \circ G$ neighborhood of $x \circ H$ neighborhood of $y \circ G \cap H = \{\}))$.

Compact subsets of Hausdorff spaces are closed.

A continuous bijection from a compact subspace to a Hausdorff space is a homeomorphism.

Examples of **Hausdorff Space** are **Metric Spaces**, and of course, **Manifolds**.

Manifold

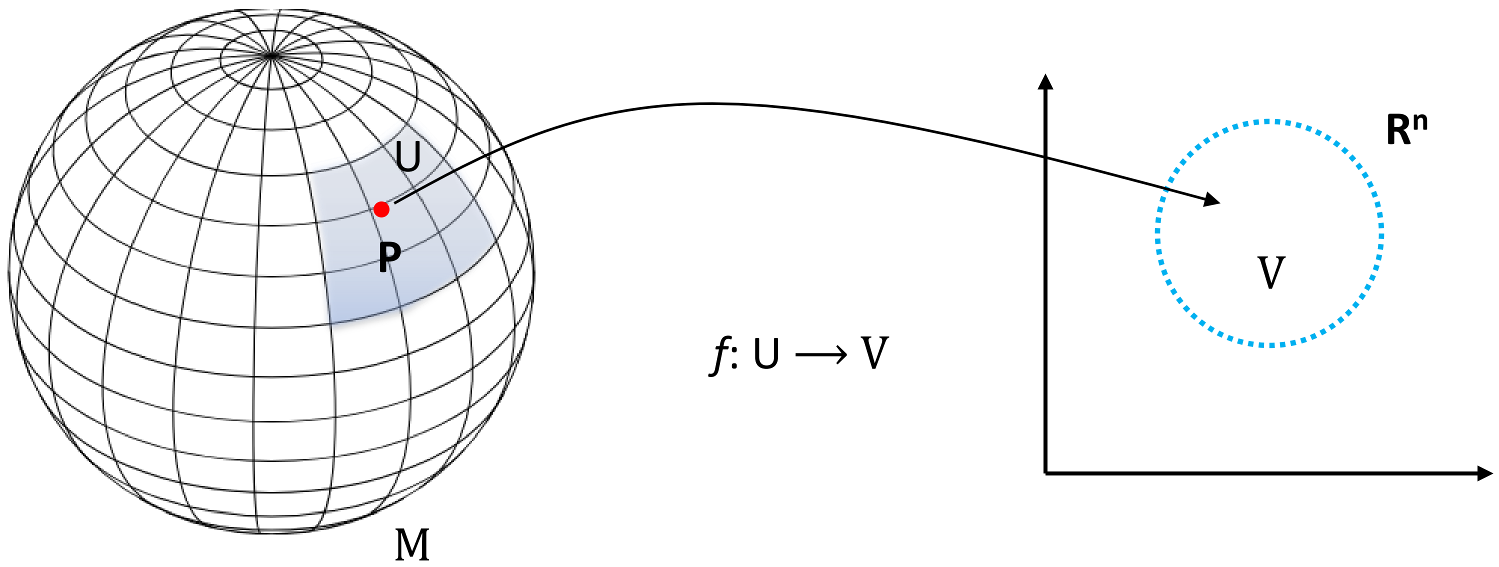
A **Manifold** is defined as a locally Euclidean Space M of dimension **n**, which is a **Hausdorff** topological space in which each point has a neighborhood homeomorphic to an open subset of $\mathbf{R}^n$.

Informally, a manifold is a topological space that locally looks like an Euclidean Space, with its topology, and as we know, a topological space is a set of points along with a set of neighborhoods for each point.

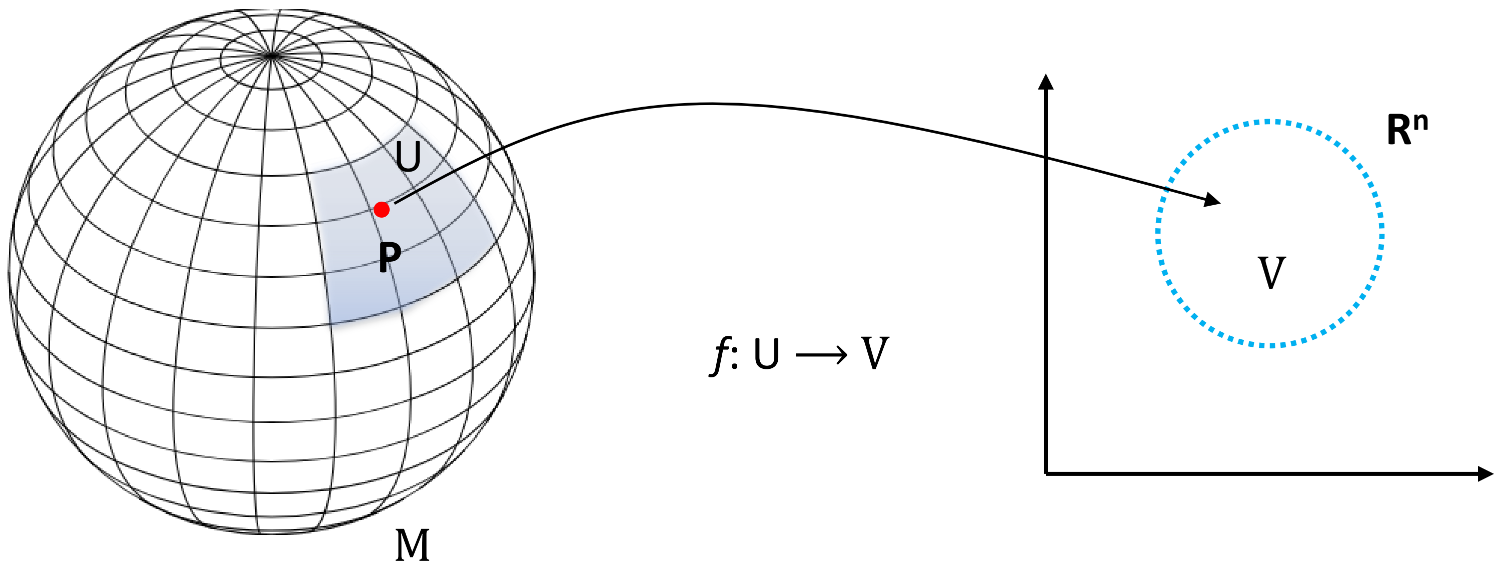
Each point in n-dimensional manifold has a neighborhood that is hemomorphic to a Euclidean Space of dimension n.

Manifold

A n-dimensional topological manifold M is a topological **Hausdorff** space with a countable base which is homeomorphic to $\mathbf{R}^n$.



Manifold

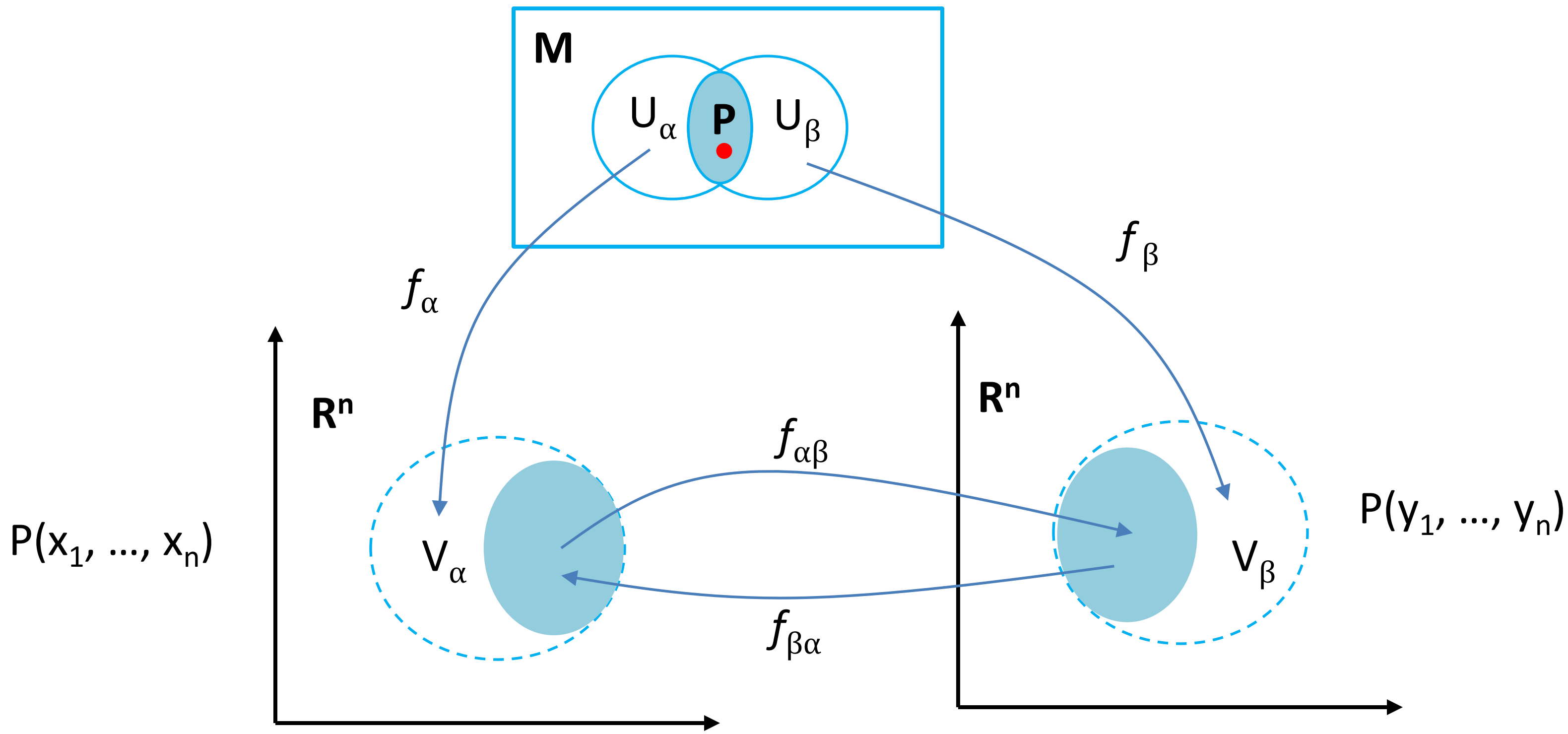


The mapping f: U ⟶ V is called a **chart** or a **coordinate system**. The set U is the domain or local coordinate neighborhood of the chart.

The image of the point $P(x_1, \ldots, x_n) \in U$ with the coordinates x_n is the point $f(P) \in R^n$ and is called coordinates of P in the chart.

A set of charts, $\{ f_\alpha \mid N \}$, with domain U_α is called an Atlas of M, the Union of $U_\alpha = M$.

Mapping of Manifold

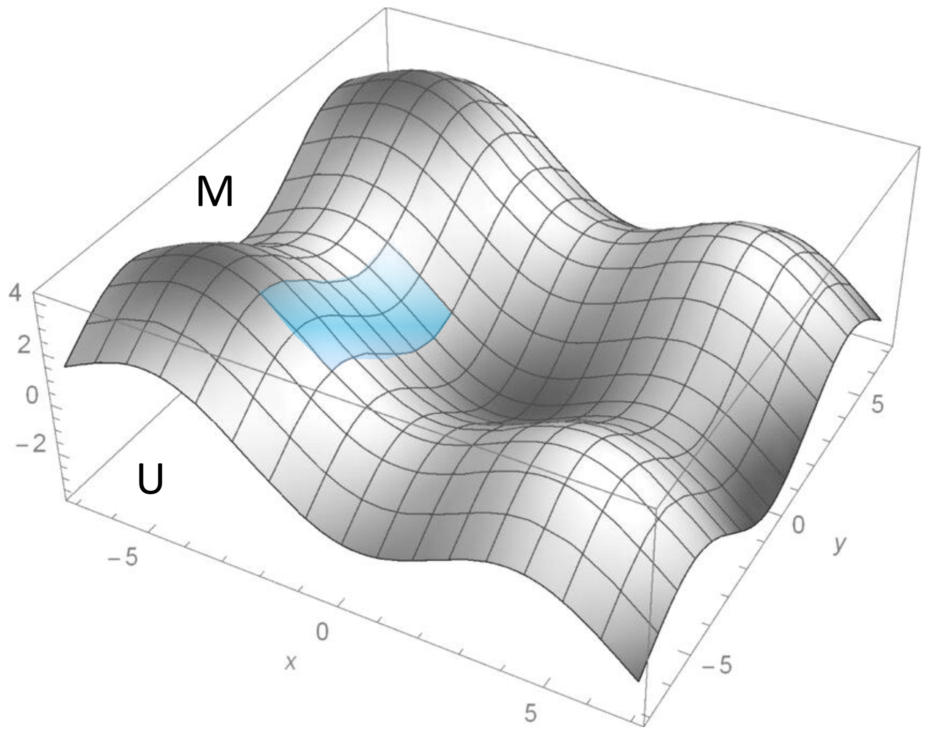


If f_α and f_β are two coordinates charts, then:

$f_{\alpha\beta} : f_\alpha (U_\alpha \cap U_\beta) \longrightarrow f_\beta (U_\alpha \cap U_\beta)$ where, $f_{\alpha\beta} = f_\alpha \circ f^{-1}{}_\beta$

$f_{\alpha\beta} : f_\alpha(P(x_1, \ldots, x_n)) \longrightarrow f_\beta (P(y_1, \ldots, y_n))$ and $f_{\beta\alpha} : f_\beta(P(y_1, \ldots, y_n)) \longrightarrow f_\alpha (P(x_1, \ldots, x_n))$

Mapping of Manifold



z = *f*(**x**)

Now, we will consider a simple Manifold M of a Graph of a function *f* such that:

$$M = f(\boldsymbol{x}) - \boldsymbol{z} = (\boldsymbol{x}, f(\boldsymbol{x})) = 0$$

Let U be the open subset of R^n and f: U ⟶ R^k be a C^∞ function. Then, the graph of function *f* is a n- sub-manifold in R^{n+k}.

$$\Gamma_f = \{(\boldsymbol{x}, f(\boldsymbol{x})) \in R^{n+k} \mid x \in U\}$$

Mapping of Manifold

We can actually think the function *f* as:

$$f: R^n \to R^{n+k} \to R^k$$

Let R^N be R^{n+k}, and let φ be a function which maps from R^N to R^k. Let **z** be a point in R^k, and **p** be a point in open set U_β in R^N, then $\mathbf{p} = \varphi^{-1}(\mathbf{z})$. Thus, if $\mathbf{z} = 0$, $\mathbf{p} = \varphi^{-1}(\mathbf{0})$, which is the kernel of φ. Thus, the **surjective** map φ is a submersion at **p**.

Again, let R^N be R^{n+k}, and let π be a function which maps from R^n to R^N. Let **x** be a point in R^n, and **p** be a point in open set U_α in R^N, then $\mathbf{p} = \pi(\mathbf{x})$. Thus, the **injective** map π is a immersion at **p**.

Thus, the intersection of $(U_\alpha \cap U_\beta)$ in R^N defines function $f = \pi \circ \varphi$. Then the graph of *f* is:

$$\Gamma_f: \boldsymbol{x} \to \left(\boldsymbol{x}, f(\boldsymbol{x})\right) \to (\boldsymbol{x}, 0)$$

However, since the intersection is $(U_\alpha \cap \{0\})$, the graph of the function *f* is a sub-manifold of n-dimension in R^{n+k}.

$$\Gamma_f: R^n \times \{0\} \to R^n$$

Differentiable Manifold

An **Atlas** defined on a Manifold M is said to be differentiable i.e., the derivatives of all order exists C^{∞}, if all of its coordinate changes are differential mappings, which are **injective** (one-to-one) and **surjective** (onto).

In other words, a differentiable manifold is a topological manifold, endowed with differentiable structure.

Differentiable manifold has the following properties:

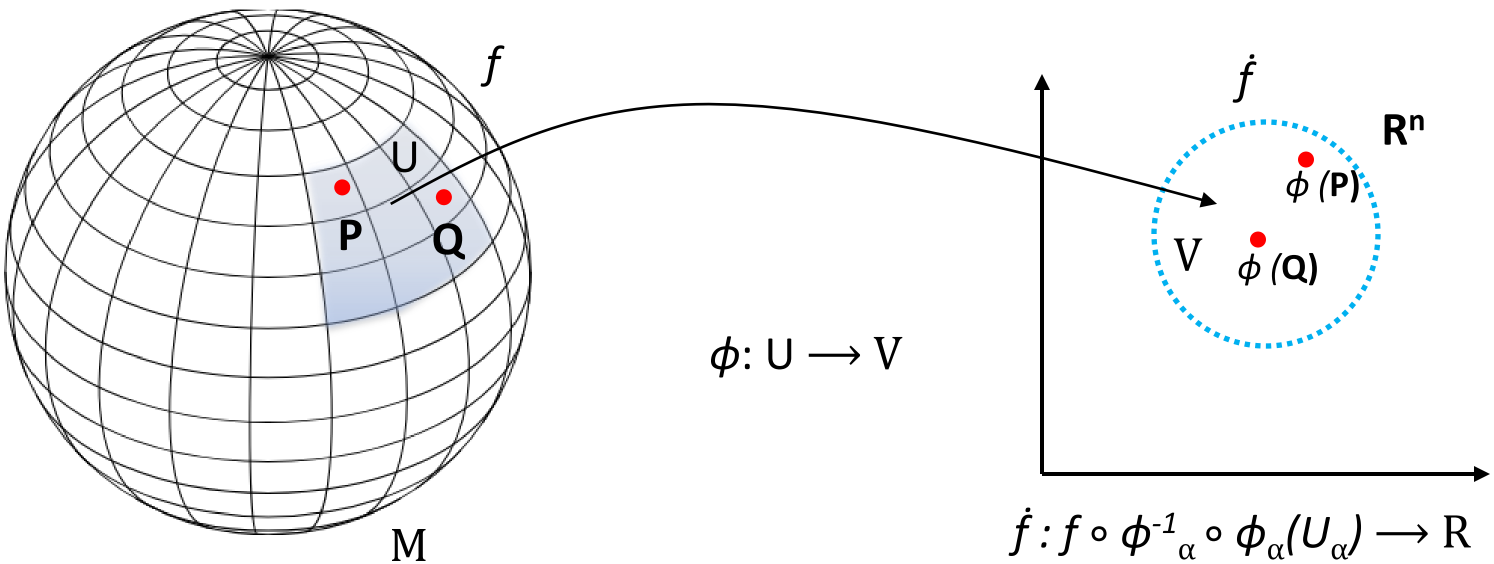
$$f_{\alpha\alpha} = Identity$$

$$f_{\beta\gamma} \circ f_{\alpha\beta} = f_{\gamma} \circ \left(f^{-1}{}_{\beta} \circ f_{\beta}\right) \circ f^{-1}{}_{\alpha} = f_{\alpha\gamma}$$

$$f_{\alpha\beta} = f^{-1}{}_{\beta\alpha}$$

Hadamard's Lemma

Let f: M $\longrightarrow$ R is a smooth function on M, let (U_α, ϕ_α) be the a Chart of some Atlas with U_α being in the domain of f. Then, $\dot{f}$: $f \circ \phi^{-1}_\alpha \circ \phi_\alpha(U_\alpha) \longrightarrow$ R is a real-valued function on R^n and f is said to be C^∞ if $\dot{f}$ is C^∞.



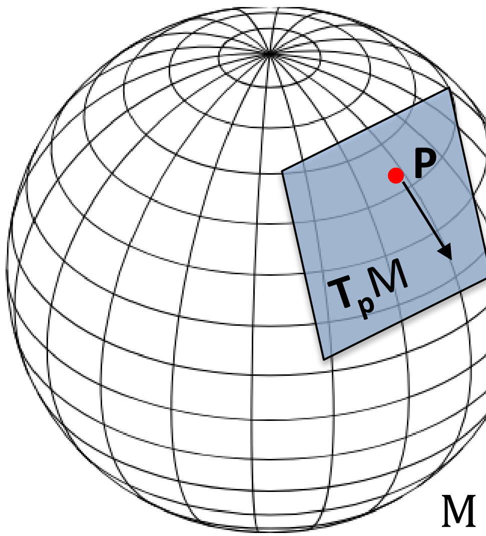
If $\phi_\alpha(P) = a_i = (a_1, \ldots, a_n)$, then in the neighborhood of a_i we can express x_i, according to Hadamard's Lemma, such that: $a_i = x_i(\phi_\alpha(P))$. If $x_i = x_i(\phi_\alpha(Q))$ and $a_i = x_i(\phi_\alpha(P))$, then

$$f(Q) = f(P) + (x_i - a_i)\partial_i f(Q)$$

Tangent Space on Manifold

As we already know, we have defined a **Tangent Space** at Point P in R^n as $\mathbf{T_p}R^n$, and the set of all tangent spaces as Tangent Bundle, **TP**.

However, now we will define a **Tangent Space on Manifold** M, which will be represented as $\mathbf{T_p}$M, and the tangent bundle will be represented as **TM**.



$$\left.\frac{\partial}{\partial x_i}\right|_P f = \left.\frac{\partial \dot{f}(x_1, \cdots, x_n)}{\partial x_i}\right|_P$$

As we know already, the basis of $\mathbf{T_p}$M is $\{ \partial / \partial \mathbf{x}_i \}$.

Tangent Bundle on Manifold

The tangent bundle will be represented as **TM**, which represents the set of all Tangent Spaces on Manifold $\mathbf{T_p}M$.

TM is the disjoint union set of all Tangent Spaces on Manifold M, such that:

$$TM = \coprod_{P_i \in M} T_{P_i} M$$

The tangent bundle TM is also the manifold.

And of course, we need to define **Affine Connection** to connect the Tangent Spaces on Manifold M as well.

Definition of Vector

Classically, a **Vector** has been represented as an **Arrow**, with **Direction** and **Magnitude**.

However, vectors can be more abstractly represented as **Derivatives** or **Linear Operators** in a more general framework, as vectors can be understood as mathematical objects with linear structures, which are invariant under transformations.

Actually, this representation would allow us to understand vectors in contexts where traditional geometric intuition might fail, especially in **topological spaces**, such as higher dimensional manifolds.

Definition of Vector

A vector at **p**, often denoted as $\mathbf{X_p}$ is defined as a derivative: a linear map $\mathbf{X_p}$: M ⟶ R, which satisfies the **Leibniz Rule**:

$$X_p(fg) = f(p)X_p(g) + X_p(f)g(p)$$

Intuitively, $\mathbf{X_p}$ measures the rate of change of *f* in the direction of vector $\mathbf{X_p}$ at point **p**.

$$X_p(f) = X_p.\nabla f$$

Let λ(t) be a smooth curve passing through a point **p**, with λ(0)=**p**. Then, the velocity vector of at point **p** acting on *f* can be defined as:

$$X_p(f) = \frac{d}{dt} f\big(\lambda(t)\big)\Big|_{t=0} = \frac{d}{dt}(f \circ \lambda)(t)\Big|_{t=0}$$

Definition of Vector

Therefore,

$$X_p(f) = X_p.\nabla f = \frac{d}{dt}(f \circ \lambda)(t) = X.\nabla f$$

Thus, in a local coordinate chart (U, φ) with the coordinates $\{x_1, \ldots, x_n\}$ around **p**, a smooth function *f* can be represented as:

$$f(x_1, \cdots, x_n)$$

Then, the derivative of *f* can be represented as:

$$\frac{\partial}{\partial x_i}(f)$$

And $\{\partial / \partial \mathbf{x}_i\}$ is the basis for the Tangent Space $\mathbf{T_pM}$ around **p**. $\mathbf{X_p}$ can be represented as:

$$X_p = \sum x^i \frac{\partial}{\partial x_i}$$

Cotangent Space on Manifold

The cotagent space at point P, denoted $T_p{}^*M$ is the dual space of the tangent space on Manifold M. It consists all the linear functions which map tangent vectors to real numbers.

$$\omega: T_pM \to R$$

If $v \in T_pM$ is a tangent vector, then is $\omega(v)$ a real number. ω is said to be **differential form**, and it is represented in Cotangent Space on Manifold M as:

$$\omega = \sum \omega_i dx^i$$

Differential forms can be understood as linear maps from the tangent space of a manifold to real number, providing a way **to measure** directional derivatives and gradients.

Dual Space

The tangent space $\mathbf{T_pM}$ and cotagent space T_p*M on a Manifold M can represent the dual space to each other.

$$df(\boldsymbol{X}) = \boldsymbol{X}(f)$$

$$df = \sum \frac{\partial f}{\partial x^i} dx^i$$

$$df(\boldsymbol{X}) = \frac{\partial f}{\partial x^i} dx^i \left(X^i \frac{\partial}{\partial x_i} \right)$$

$$= X^i \frac{\partial f}{\partial x^i}$$

$$= \boldsymbol{X}(f)$$

Differential 2-Form

As we already know, the differential 2-Form can be represented as:

$$\omega = dx^i \wedge dx^j$$

Differential 2-Form is a mathematical object, which takes two tangent vectors as input, and produce a scalar as output. 2-Form is **anti-symmetric**, meaning if two input vectors are swap, the sign of the output changes.

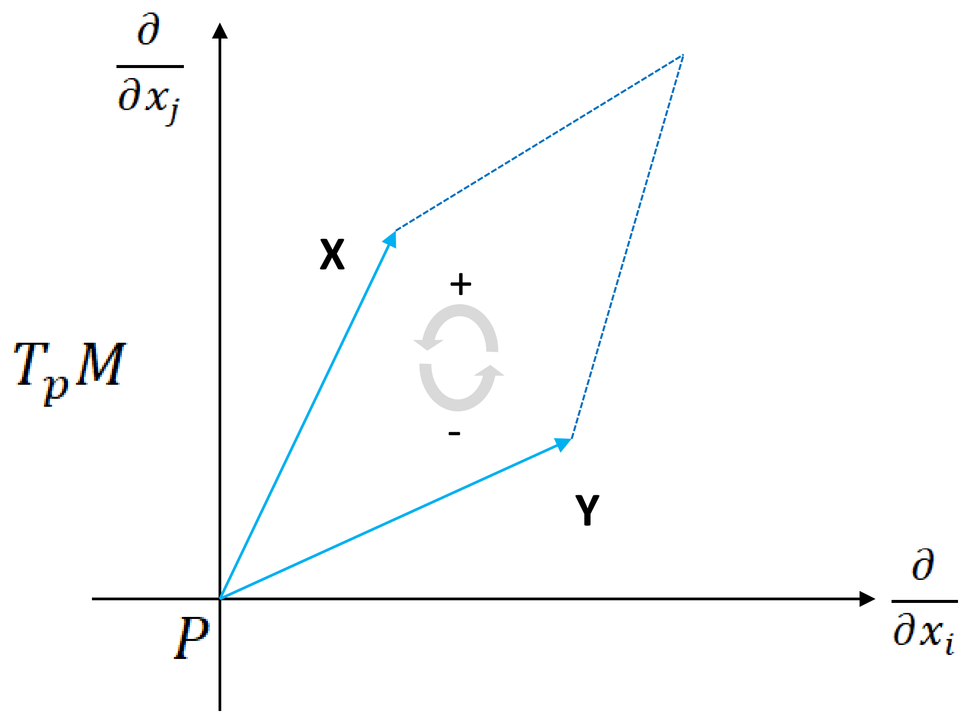
Formally, 2-Form is **skew-symmetric bilinear** map which takes 2 vectors and returns a scalar.

$$\omega_p\colon T_pM \times T_pM \to R$$

$$\omega_p(X, Y) = -\omega_p(Y, X)$$

Differential 2-Form

Given a 2-form $\omega = dx^i \wedge dx^j$, we can evaluate two tangent vectors, **X** and **Y** such that:



As we can see, $\omega = dx^i \wedge dx^j$ defines the signed **Area of Parallelogram**, spanned by **X** and **Y**.

$$dx^i \wedge dx^j(X, Y) = dx^i(X)dx^j(Y) - dx^j(X)dx^i(Y)$$

Differential 3-Form

Likewise, the differential 3-Form can be represented as:

$$\omega = dx^i \wedge dx^j \wedge dx^k$$

Differential 3-Form would take three tangent vectors as input, and produce a scalar as output.

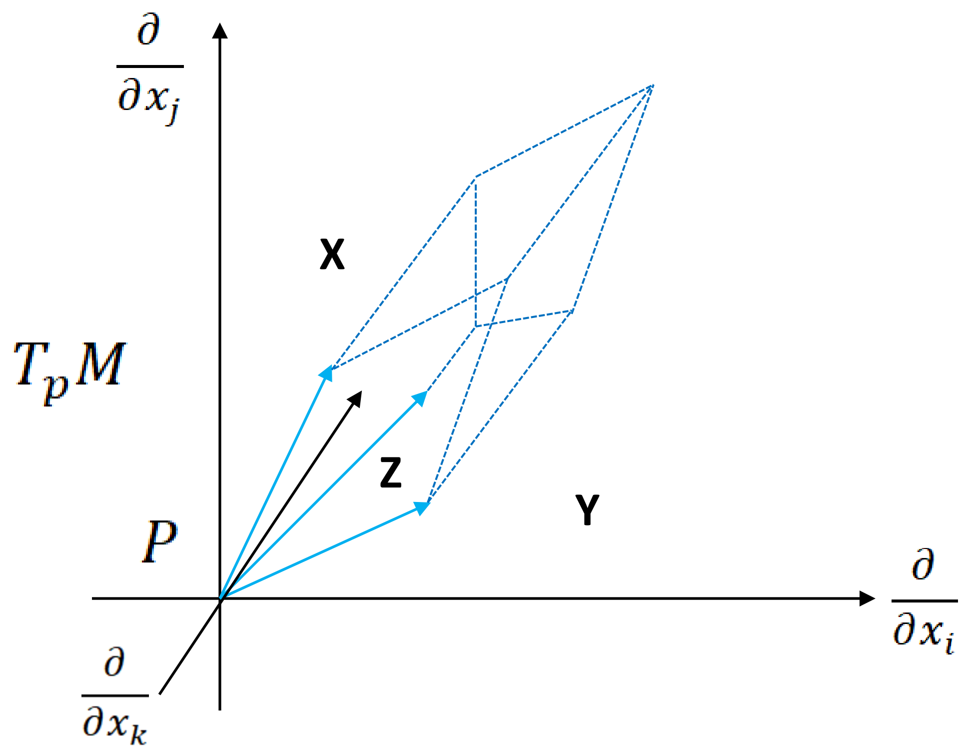
$$\omega_p : T_pM \times T_pM \times T_pM \to R$$

$$\omega_p(X, Y, Z) = -\omega_p(Y, X, Z) = -\omega_p(X, Z, Y)$$

$$dx^i \wedge dx^j \wedge dx^k = \begin{vmatrix} dx^i(X) & dx^i(Y) & dx^i(Z) \\ dx^j(X) & dx^j(Y) & dx^j(Z) \\ dx^k(X) & dx^k(Y) & dx^k(Z) \end{vmatrix}$$

Differential 3-Form

Given a 3-form $\omega = dx^i \wedge dx^j \wedge dx^k$, we can evaluate two tangent vectors, **X**, **Y** and **Z** such that:



As we can see, $\omega = dx^i \wedge dx^j \wedge dx^k$ defines the signed **Volume of Parallelepiped**, spanned by **X**, **Y**, **Z**.

Differential K-Form

As we know already, we can extend this to general differential form of k terms, which is simply called Differential K-Form.

There are ${}^{n}P_{k}$: k permutations of n elements in Differential k-Form on an Manifold M with n Dimension, as k <= n:

$$ {}^{n}P_{k} = \binom{n}{k} $$

$$ dx_{I_p} = (dx_1)_p \wedge \cdots \wedge (dx_n)_p $$

Exterior Product

Given two differential forms ω and η of degree p and q respectively, the **Exterior Product** of ω and η, is a differential form of degree (p + q). The Exterior Product, also known as **Wedge Product**, is **bilinear**, **antisymmetric** and **associative**.

$$\omega \wedge \eta = \underbrace{T_pM \times \cdots \times T_pM}_{\text{p + q}} \to R$$

$$\omega \wedge \eta(X_1, \cdots, X_n) = \frac{1}{p!\,q!} \sum_{\sigma \in S_{p+q}} sign(\sigma)\omega(X_{\sigma(1)}, \cdots, X_{\sigma(p)})\eta(X_{\sigma(1)}, \cdots, X_{\sigma(q)})$$

S_{p+q} is a Symmetric Group on p + q elements, and sign(σ) is the sign of permutation. The wedge product has the following properties.

$$\omega \wedge \eta = (-1)^{pq} \eta \wedge \omega$$

Orientable Manifold

Let M be an n-manifold. An orientation of M is a rule for assigning to each p ∈ M an orientation of T_pM.

An orientation of a manifold M is smooth, or simply $C\infty$, if for every p ∈ M, there exists a neighborhood U of p and an no-vanishing n-form $\omega \in \Omega^n(U)$ with the property:

$$\omega_p \in \Lambda^n\left(T_p^{\ *}M\right)_+$$

$$\Lambda^n\left(T_p^{\ *}M\right)_+ = \Lambda^n\left(T_p^{\ *}M\right) \setminus \{0\}$$

If ω has this property, we call ω as **Volume Form**, when the degree of differential form is equal to the dimension of differentiable manifold. A manifold admits a **no-vanishing volume form** if and only if it is **orientable**. An orientable manifold has infinitely many volume forms.

By switching the label, we will call **Reverse Orientation** of M.

Hodge Star Operator

Let M be an orientable n-manifold, especially Orientiable Riemannian Manifold, then Hodge Star Operator will allow us to transform k-form to (n-k) form. Hodge Star Operator is an isomorphic linear map between the space of k-form and (n-k) form.

$$*: \Lambda^k\left(T_p{}^*M\right) \rightarrow \Lambda^{n-k}\left(T_p{}^*M\right)$$

If (M, g) be an Orientiable Riemannian Manifold n-manifold, with k-form ω, *ω of (n-k) form is represented as the association of **Volume Element** with the Inner Product of the Differential Forms: $\omega \in \Lambda^k(T_p{}^*M)$ and $\eta \in \Lambda^{n-k}(T_p{}^*M)$:

$$*\,\omega = \omega \wedge^* \eta = \langle \omega, \eta \rangle dV_M$$

$$dV_M = \sqrt{\det(g)}\, dx^1 \wedge \cdots \wedge dx^n$$

dV_M is the Volume Form of the Orientiable Riemannian Manifold, and *det*(g) is the determinant of the Metric Tensor associated with the Riemannian Manifold M.

Hodge Star Operator

Let ω be:

$$\omega = \sum_{i_1 \cdots i_k} \omega_{i_1 \cdots i_k} dx^{i_1} \wedge \cdots \wedge dx^{i_k}$$

Then, *ω can be represented as

$$* \omega = \frac{\sqrt{|g|}}{(n-k)!} \sum_{j_1 \cdots j_{n-k}} \varepsilon^{i_1 \cdots i_k j_1 \cdots j_{n-k}} \omega_{i_1 \cdots i_k} dx^{j_1} \wedge \cdots \wedge dx^{j_{n-k}}$$

As we can see Hodge Star Operator allow us to transform k-form space to its dual (n-k) form space, such as transforming a line elements to area elements or vice versa.

Pullback on Differential k-Form

Let f: M ⟶ N be a map between Manifolds M and N, and let ω be the a k-form on N. Then, the pullback of the k-form ω by f is denoted as f^* ω, and is defined as a k-form on M.

$$f: M \rightarrow N$$

$$df_p: T_pM \rightarrow T_{f(p)}N$$

For any point p ∈ M, and a vector v ∈ T_pM, it would give as 1-Form,

$$(f^*\omega)_p(v) = \omega_{f(\boldsymbol{p})}(f_*v) = \omega_{f(\boldsymbol{p})}(df_p(v))$$

$$\omega_{f(\boldsymbol{p})}\left(df_p(v)\right) = \omega_j(f(p))\frac{\partial f^i}{\partial x_i}dx^i$$

Pullback on Form has the following properties:

$$f^*(d\omega) = d(f^*\omega)$$

Pullback on Differential k-Form

Then, for any point p ∈ M, and vectors $v_1, \ldots, v_k \in T_pM$:

$$(f^*\omega)_p(v_1, \cdots, v_k) = \omega_{f(\boldsymbol{p})}(df_p(v_1), \cdots, df_p(v_k))$$

If ω is defined on N such that:

$$\omega = \mu(y)dy^{\mathrm{j}_1} \wedge \cdots \wedge dy^{\mathrm{j}_\mathrm{k}}$$

Then,

$$(f^*\omega)_p = \mu(f(x))f^*\left(dy^{\mathrm{j}_1}\right) \wedge \cdots \wedge f^*\left(dy^{\mathrm{j}_\mathrm{k}}\right)$$

As we can see, the pullback on manifold defines how a differential form in manifold **N** can be represented in terms of k-form in manifold **M**.

Pullback on Volume Form

Now, let U and V be open subsets of $\mathbf{R}^n$ and, let μ be a volume form:

$$\mu = h(y)dy^1 \wedge \cdots \wedge dy^{\mathrm{n}}$$

Then,

$$f^*(\mu) = (h \circ f)\det\left(\frac{\partial f^i}{\partial x^{\mathrm{j}}}\right)dx^1 \wedge \cdots \wedge dx^{\mathrm{n}}$$

$$\det\left(\frac{\partial f^i}{\partial x^{\mathrm{j}}}\right) = \det(J)$$

If *det(J)* > 0, the map *f* **expands** the volume, and if *det(J)* < 0, the map *f* **compresses** the volume, but if *det(J)* = 0, the map f **collapses** the volume.